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Hierarchical optimization of virtual power plants via sequential Game-Based Multi-Agent reinforcement learning

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ABSTRACT

The integration of distributed energy resources (DERs) in virtual power plants (VPPs) is a critical enabler of low-carbon power system transformation; however, market uncertainty and conflicts among multiple decision-making agents pose major challenges for efficient scheduling. To address these issues, this paper proposes a hierarchical multi-agent reinforcement learning framework based on a hybrid-strategy sequential game. A novel dynamic role reduction mechanism is established for both supply and demand sides, enabling the aggregation of large-scale and heterogeneous VPP resources into a compact dual-agent sequential game representation. The framework employs a two-layer architecture. At the upper layer, hybrid pricing strategies are optimized through a sequential game where the supply side acts as the leader to maximize generation profits, and the demand side serves as the follower by adaptively adjusting load levels, achieving a robust subgame perfect Nash equilibrium despite conflicting objectives. In the lower layer, the scheduling task is formulated as a constrained Markov game, with Lagrangian relaxation incorporated to handle power balance, temporal coupling, and resource competition constraints, while quantifying both exogenous and endogenous uncertainties for adaptability in real-time market environments. We introduce the Sequential Game-based Multi-Agent Deep Deterministic Policy Gradient (SG-MADDPG) algorithm, which integrates the hierarchical game into an enhanced Multi-Agent Deep Deterministic Policy Gradient (MADDPG) framework. This algorithm features a graph neural network-enhanced adversarial dual-path critic to capture asymmetric supply–demand interactions and mitigate environmental non-stationarity, alongside a distributionally robust optimization (DRO) mechanism to improve training stability and policy robustness. Simulation results on a real-world VPP case demonstrate that the proposed approach increases overall profits by 14.7%-25.3%, reduces default risk by approximately 65%, and enhances deployable operational profits by more than 20%. Compared to centralized model predictive control (MPC) and traditional Stackelberg game-based methods, our framework exhibits superior performance, confirming its effectiveness and applicability in uncertain electricity market environments.

1. Introduction

1.1. Research Background and Significance

Under the global consensus of the "Dual Carbon" goals, the power system is transitioning from traditional fossil fuel dominance to clean energy. As an emerging energy aggregation model (Zhou et al., 2025; Lin et al., 2023; Yao et al., 2024; Yang et al., 2025; Meng et al., 2025), the virtual power plant (VPP) integrates heterogeneous distributed energy resources (DERs)—such as distributed photovoltaic (PV), wind power, energy storage systems, and controllable loads—into a coordinated virtual entity through advanced information and communication technologies and control algorithms. This enables the unified dispatch and optimization of heterogeneous resources (Xu et al., 2024; Lin et al., 2025) and has emerged as a key facilitator of the low-carbon transition (Liu et al., 2025; Liu et al., 2024; Kim and Kim, 2023).

Traditional VPP scheduling optimization methods emphasize mathematical modeling and solution efficiency, realizing power allocation, frequency regulation, and cost minimization through precise physical or economic models (Jendoubi and Bouffard, 2023; Zhang et al., 2021). For example, in Ref. (Guo et al., 2025) proposes a collaborative power control strategy for VPPs based on multi-agent systems that employs consensus algorithms to defend against cyberattacks, homomorphic encryption to protect privacy, and an event-triggered mechanism to

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reduce the communication burden. Nevertheless, this approach neglects electricity market uncertainty and multi-stakeholder conflicts of interest. In Ref. (Song et al., 2025) developed a mixed-integer nonlinear programming (MINLP) model to optimize the operational strategy of a wind-solar-thermal generation system assisted by shared energy storage in joint electricity-carbon markets. However, this method relies heavily on accurate physical models and exhibits limited adaptability to real-time uncertainty. Similarly, Ref. (Thangavelu et al., 2025) constructed an adaptive AI energy system model for a cryo-polygeneration plant (CPP) using adaptive AI to predict the distributed energy performance and optimize multi-energy coordination. Its limitation lies in its high computational complexity, particularly during the AI adaptive update phase.

Game-theoretic methods focus on multi-agent interactions and equilibrium solutions, modeling strategic interactions among prosumers, operators, and the grid to achieve Nash equilibrium or Stackelberg game equilibrium for optimizing resource allocation and efficiency of the markets. In Ref. (Alam et al., 2024) introduced a blockchain-based peer-to-peer energy trading framework, utilizing Mixed Integer Linear Programming (MILP) to optimize prosumer profits and employing Ethereum smart contracts to achieve decentralized coordination of DERs. Results indicate that this framework enhances grid stability and renewable energy utilization. However, the method does not adequately account for the dynamic process of sequential games and real-world market uncertainties. Ref. (Zhou et al., 2023) proposed a bi-level optimization model for an urban VPP, where the lower level considers user satisfaction based on Incentive-Based Demand Response (IBDR), and the upper level optimizes generation dispatch and market transactions. Results show that the model supports supply–demand balance and minimizes costs. Nevertheless, it assumes a complete information game, thereby neglecting mixed strategies under uncertain environments. In (Yin and Ye, 2025) a distributed multi-objective African Vulture Optimization Algorithm(AVOA), combined with deep neural networks to accelerate optimization. Tests on 118-bus and 1790-bus systems, this algorithm reduced carbon emissions by 3.5%-9.95% and costs by 2.16%-4.8%. Its limitation lies in the strong static assumptions and weak adaptability to dynamic uncertainties.

Reinforcement learning (RL) methods emphasize adaptation to uncertainty and learning optimal policies through agent-environment interactions to achieve robust scheduling in dynamic environments. In Ref. (Häberle et al., 2023) proposed hierarchical multi-agent deep reinforcement learning (HMADRL), which adopts a centralized training with decentralized execution (CTDE) paradigm to handle mutual decision exclusivity and introduces a hot experience pool to reduce exploration. Simulations show the improved reliability of multi-energy multi-microgrid systems; however, the absence of game-equilibrium guidance leads to unstable policies in multi-agent, non-stationary environments. Ref. (Zhang et al., 2025) developed an active distribution network scheduling method based on a safe reinforcement learning framework to ensure compliance with distribution network constraints. Comparisons demonstrate superiority over existing safe reinforcement learning (SRL) and classical optimization methods; however, they neglect multi-agent game non-stationarity and fail to handle dynamic inter-agent conflicts effectively. In Ref. (Xu et al., 2024) proposed a robust preventive-corrective security-constrained Optimal Power Flow (OPF) model and addressed VPP faults using a Lagrangian-based soft actor (L-SAC) algorithm. Tests on IEEE systems show improved computational efficiency and reliability; however, they do not consider sequential games under mixed strategies and struggle with multi-round interactions under uncertainty.

Recent advancements in Deep Reinforcement Learning (DRL) have further stimulated the practical application of machine learning methods in energy systems. In Ref. (Roy and Bhaduri, 2023) developed a Dense SPH-YOLOv5 model, which integrates a DenseNet module into the backbone network to enhance the retention and reuse of critical feature information. Additionally, extra feature fusion layers and a swin-Transformer prediction head (SPH) were incorporated, leveraging advanced self-attention mechanisms to reduce the algorithm's computational complexity. Ref. (Jiang et al., 2022) introduced a semi-supervised learning framework based on a multi-graph cooperative neural network, improving graph representation by integrating complementary multi-graph information, achieving classification accuracy 5–10% higher than baselines on multi-view datasets. The research on such advanced DRL models has demonstrated significant effectiveness for existing energy applications. In energy-specific applications, Ref. (Guo et al., 2025) proposed an evolutionary DRL algorithm, Cross-Entropy Method (CEM)-MADDPG, for electric vehicle thermal management, which enhances robustness through cross-entropy and reduces energy consumption by 19.8% compared to rule-based methods. Similarly, Ref. (Li et al., 2025) applied the Deep Deterministic Policy Gradient (DDPG) algorithm from DRL for the multi-objective optimization of an integrated electricity-gas-heat energy system, improving the reward value by 6.09% over proximal policy optimization (PPO), thereby verifying reductions in cost and emissions. These DRL advancements motivate the integration of cutting-edge DL techniques—such as attention mechanisms, graph cooperation, and evolutionary reinforcement—to enhance the practicality of machine learning in VPP optimization by addressing noisy data, multi-view representations, and multi-objective robustness.

To sum up, traditional optimization techniques are heavily biased towards model fidelity, game theoretical techniques are frequently biased towards restrictive static assumptions, and RL techniques are susceptible to non-stationary environments. Although recent DRL developments have made significant improvements in the feature extraction ability and robustness in aspects such as damage detection and energy system optimization, they have not been successfully extended to tackle sequential game dynamics in large-scale VPPs (Guan et al., 2025; Wei et al., 2025; Srivastava et al., 2024; Upadhayay et al., 2024; Xu Q, Su Z, Li P, et al. Cooperative Energy Provisioning Services with Virtual Power Plants in Smart Grid Internet of Things: A Coalition-Stackelberg Game Approach[J]. IEEE Internet of Things Journal, 2025). Existing methods tend to work well under deterministic scenarios, but show significant limitations when faced with uncertainty, heavy computational load and scalability issues, which constrain their use to large-scale, dynamic VPP scenarios. Moreover, these approaches have problems in solving collaborative conflicts between multiple agents (Zhang et al., 2025; Liu et al., 2022; Fan et al., 2025; Lu et al., 2023; Fan et al., 2023). Overall, most previous research takes a single methodological view and comprehensive frameworks that closely integrate game theoretic modeling with learning-based decision making are still lacking.

1.2. Research gaps and contributions

In summary, although there have been some significant developments in the literature, there are still some critical gaps. Existing studies often overlook the role of mixed-strategy sequential games in uncertain electricity market environments, where multi-agent approaches are often limited to pure strategies or synchronous decision-making. This limitation makes it impossible to accurately model dynamic interactions between supply-side leaders and demand-side followers, leading to deviations from globally optimal solutions (Dai et al., 2025; Chen et al., 2022; Golpîra and Marinescu, 2024; Liu et al., 2023; Shao et al., 2024). Also, most research assumes a fixed agent population or only considers single-level optimization and does not capture strategic interactions across multiple hierarchical levels. Multi-Agent RL (MARL) techniques also face serious challenges such as environmental non-stationarity and training instability (Zhang et al., 2025; Zhou and Preindl, 2023). Standard algorithms, such as Multi-Agent Deep Deterministic Policy Gradient (MADDPG), can converge to local optima, and do not exploit the full benefits of Nash equilibrium guidance or representational benefits of Graph Neural Networks (GNNs) (Chen et al., 2024; Wu et al., 2024).

To address these gaps, a bi-level optimization framework based on mixed-strategy sequential games is proposed within a MARL paradigm. The main contributions of this study are summarized as follows:

1) Hierarchical Hybrid-Strategy Framework for Large-Scale VPPs: This study addresses the shortcomings of traditional VPP scheduling methods in coordinating large-scale DERs and resolving multi-agent conflicts by proposing a hierarchical optimization framework grounded on hybrid-strategy sequential games and MARL. A novel dynamic role-reduction mechanism is proposed to aggregate heterogeneous resources into two agent types, the supply side and the load side. These agents interact non-cooperatively through the electricity price signals. At the upper level, the supply-side agents play the role of leaders to maximize profit through pricing strategies, and the load-side agents play the role of followers to dynamically adjust demand. This interaction leads the system to a subgame-perfect Nash equilibrium and takes into account multi-party conflicts. Unlike the traditional Stackelberg formulations, the proposed approach eliminates assumptions of pure strategies and perfect information, which greatly enhances the scalability and adaptability to scenarios with tens of thousands of aggregated resources.

2) Constrained Markov Game and DRO for Lower-Level Scheduling: The lower-level scheduling is formulated as a constrained Markov game, with constraints such as power balance, temporal coupling, and resource competition handled via Lagrangian relaxation. Market disturbances and internal–external uncertainties are explicitly considered. By embedding this game into an improved MADDPG algorithm, we design a Nash equilibrium-oriented differentiated reward function, which guides multi-agent policies to converge to a Nash equilibrium while ensuring consistent expected returns across different pure strategies. This design effectively mitigates the poor policy convergence of traditional MARL in non-stationary environments.

3) Robustness to Non-Stationarity and Market Uncertainty: To further address environmental non-stationarity and market uncertainty, this study incorporates a Graph Neural Network-enhanced Implicit Adversarial Dual-path Critic (GIADC) structure at the upper level to implicitly model the asymmetric dynamic interactions between supply and demand, thereby decoupling the impact of environmental variations. At the lower level, a Distributionally Robust Optimization (DRO) mechanism is integrated to enhance robustness against model errors and volatile market conditions.

Through these innovations, the proposed framework demonstrates distinct advantages in coordinating heterogeneous resources, managing non-cooperative game interactions, and coping with market uncertainties in VPPs. It effectively overcomes the limitations of traditional centralized scheduling, classical Stackelberg games, and basic MARL approaches in terms of scalability, multi-agent non-stationarity, and policy convergence, offering an efficient and reliable solution for optimal VPP scheduling in large-scale, dynamic market environments. Experimental results show that compared with centralized MPC optimization and traditional Stackelberg benchmark, it improves global profit by 14.7%-25.3%, reduces default risk by 65%, and increases profit by more than 20% in actual deployment.

2. MARL two-layer scheduling decision framework

2.1. VPP distributed resource characteristics analysis

A VPP aggregates various resources distributed across different locations using information technology to form a virtual, unified "power plant." Through intelligent coordination and optimization, it achieves centralized control and scheduling of distributed resources, thereby participating in the electricity market, grid balancing, and energy trading as a single entity. The characteristics of the energy resources in a VPP are analyzed as follows:

1) Distributed PV: As an important clean energy source in a new type of power system, PV use solar radiation as the primary energy source, offering advantages such as renewability, low carbon emissions, and local consumption. Their output power is affected by light intensity, meteorological conditions, and installation angle, which exhibit intermittency and volatility. This makes it difficult to achieve a continuously rated power output, posing challenges to the stable operation of the power grid and power balance. Therefore, it is necessary to combine energy storage and smart grid technologies to mitigate this instability. Distributed PV is categorized as the "supply side." The mathematical model is formulated as follows:

$$P_t^{pv} = P_{pv-T}\frac{L(t)}{L_{STC}}\left[f_{pv}\left(T(t) - T_{STC} + 30\cdot\frac{L(t)}{L_{STC}}\right) + 1\right] \tag{1}$$

where $Ppv\ t$ is the photovoltaic power at time t; $P_{pv\text{-}T}$ is the rated output power of ($T_{STC} = 25\circ\mathrm{C}$, $L_{STC} = 1000\ \mathrm{W/m^2}$) under standard test conditions; f_{pv} is the temperature coefficient of the output power; $T(t)$ is the ambient temperature; and $L(t)$ is the solar irradiance. The PV output power range satisfies the following constraints:$P^t_{pv,\min} \leqslant P^t_{pv} \leqslant P^t_{pv,\max}$.

2) Industrial load: Industrial load refers to the electrical energy consumed to drive the operation of industrial equipment. Electricity demand can be adjusted by controlling the production equipment and rearranging production schedules, thereby achieving energy savings and efficiency improvements. Specifically, this behavior manifests in two ways: active load reduction during grid-peak periods to conserve energy and intentional load increase during valley periods to maintain production levels, thereby lowering overall electricity costs. Accordingly, the industrial load is defined as the "load side." For flexible shiftable industrial loads, the electricity market adopts a subsidy-based pricing strategy for energy scheduling. The mathematical model of load shifting based on the subsidy price is as follows:

$$\begin{cases} P^{Max,t}_{L,dem,Mov} = \left[\dfrac{1}{\sqrt{2\pi}\,\theta} exp\left(-\dfrac{(\pi_t - \partial)^2}{2\theta^2}\right)\right]\cdot P^t_{L,dem} \\ \partial = 0.25\cdot\dfrac{\pi_t - \pi_{t,\min}}{\pi_{t,\max} - \pi_{t,\min}} + 0.25, \theta = 0.05 \end{cases} \tag{2}$$

where $PMax,t\ L,dem,Mov$ represents the maximum load transfer power of the industrial load in the VPP at time t within the scheduling cycle; $Pt\ L,dem$ is the actual power consumption of the industrial load; π_t is the subsidy electricity price given by the grid; $\pi_{t,\min}$ and $\pi_{t,\max}$ represent the minimum and maximum values of the subsidy electricity price, respectively.

For a flexible reducible industrial load, the maximum allowable load reduction at any given moment is closely related to the capacities of PV production, EVs, energy storage systems, and the grid-connected scheduling center within the VPP at that specific moment. The upper limits of the flexible, reducible load and negative spinning reserve capacity were calculated as follows:

$$\begin{cases} \sum_{L,dem} R^{Down,t}_{Cut} \geqslant c^{down}_{pv}\cdot P^t_{pv} + P^t_{ESS,out} + P^t_{EV,i,out} + P^t_{sch_toL} \\ P^{Max,t}_{L,dem,Cut} = c_{Cut}\cdot R^{Down,t}_{Cut} \end{cases} \tag{3}$$

where $Rdown,t\ Cut$ is the negative spinning reserve capacity of the industrial load, $cdown\ pv$ is the negative spinning reserve capacity coefficient of the distributed PVs for handling load forecasting deviations, $PMax,t\ L,dem,_{Cut}$ is the maximum load reduction power that the industrial load can execute in the VPP, and c_{Cut} is the flexible load reduction coefficient.

Constraints of load transfer:

$$\begin{cases} -P^{Max,t}_{L,dem,Mov} \leqslant P^t_{L,dem,Mov} \leqslant P^{Max,t}_{L,dem,Mov}, t \in [1, T] \\ \int_1^T P^t_{L,dem,Mov} dt = 0 \end{cases} \tag{4}$$

where the first equation indicates that the transfer power (positive for

upward adjustment and negative for downward adjustment) at any time is subject to upper and lower bound constraints. The second equation is the energy conservation constraint, which means that 'the electricity shifted away during the daytime must be compensated at night' with the net adjustment amount being zero over the entire optimization horizon.

Constraints for load reduction:

$$-P^{Max,t}_{L,dem,Cut} \leqslant P^{t}_{L,dem,Cut} \leqslant P^{Max,t}_{L,dem,Cut}, t \in [1, T] \tag{5}$$

where $\sum_t P^t_{L,dem,Cut}\Delta t \leqslant E^{Max}_{Cut}$ serves as the limit for the cumulative reducible energy that can be tolerated. *PMax,t L,dem,Mov* is the upper limit of the flexible load transfer amount calculated using Equation (3), *PMax,t L,dem,Cut* is the upper limit of the flexible load reduction amount calculated using Equation (4).

3) Energy storage unit: The energy storage unit refers to various dispersed household energy storage systems or small-scale dedicated energy storage devices. To improve scheduling efficiency, energy storage units within a certain area can be aggregated through the network and treated as a unified aggregated scheduling resource. When an energy storage unit supplies power to the grid, it can be regarded as the "supply side;" when it replenishes energy, it can be regarded as the "load side. " The energy storage unit achieves revenue maximization by storing energy during low-price periods and releasing it during high-price periods. The power output/charging state of the energy storage unit relies on the state of charge (SOC), rated capacity, and charging/discharging efficiency. The dynamic input/output is expressed as follows:

$$P^t_{ESS} = \begin{cases} P^t_{ESS,out}(\text{Discharge}, P > 0) \\ P^t_{ESS,in}(\text{Charge}, P < 0) \end{cases} \tag{6}$$

where *Pt ESS,out* and *Pt ESS,in* are the charging and discharging powers of the energy storage unit, respectively.

The battery state at the current moment is updated based on the battery state at the previous moment and the charging/discharging situation within the current time step. Therefore, the variation in the SOC of the energy storage unit over time is expressed as follows:

$$SOC^t_{state} = SOC^{t-1}_{state}\lambda_{soc} + \frac{\left(P^t_{ESS,in}\eta_{ESS} - \frac{1}{\eta_{ESS}}P^t_{ESS,out}\right)\Delta t}{E_{ESS}} \tag{7}$$

where *SOCt state* represents the battery SOC at the current time *t*; λ_{soc} is a regulation factor that affects the sensitivity of SOC variation; and η_{ESS} represents the efficiency of the energy storage system.

$$\begin{cases} P^t_{ESS,in,\max} \leqslant P^t_{ESS} \leqslant P^t_{ESS,out,\max} \\ SOC_{\min} \leqslant SOC^t_{state} \leqslant SOC_{\max} \end{cases} \tag{8}$$

4) EVs: The battery capacity of a typical household EV ranges from 15 kWh to 100 kWh. By aggregating EVs, a VPP can treat them as flexible power sources to inject electricity into the grid to support peak demand or as flexible loads to consume surplus electricity, thereby achieving a demand response. Therefore, when an electric vehicle supplies power to the grid, it is defined as the "supply side;" when it is charging, it is defined as the "load side.".

Its dynamic power input/output is expressed as follows:

$$P^t_{EV,i} = \begin{cases} P^t_{EV,i,out}(\text{Discharge}, P > 0) \\ P^t_{EV,i,in}(\text{Charge}, P < 0) \end{cases} \tag{9}$$

where *Pt EV,i,out* and *Pt EV,i,in* are the charging and discharging powers of the EV, respectively.

The energy renewal equation for EV batteries is:

$$SOC^t_{EV,i} = \begin{cases} SOC^{t-1}_{EV,i} + \dfrac{\eta_{in}P^t_{EV,i,out}\Delta t}{Q_e}(\text{Charge}) \\ SOC^t_{EV,i} + \dfrac{P^t_{EV,i,in}\Delta t}{\eta_{out}Q_e}\ (\text{Discharge}) \end{cases} \tag{10}$$

In the formula, *SOCt EV,i* represents the remaining energy of the battery of the *i* EV at time *t*. η_{in} and η_{out} represent the charge/discharge efficiency. Q_e represents the rated capacity of the battery.

The remaining capacity of EV batteries and their charging/discharging must satisfy the upper and lower limits to avoid overcharging/discharging.

$$\begin{cases} E^t_{EV,i,\min} \leqslant E^t_{EV,i} \leqslant E^t_{EV,i,\max} \\ P^t_{EV,in,\max} \leqslant P^t_{EV,i} \leqslant P^t_{EV,out,\max} \end{cases} \tag{11}$$

5) VPP grid connection interface with external power grid: At the physical connection level, individual distributed resources of the VPP are electrically connected to the distribution network through their own access devices, and the connection point is the traditional "grid connection point" or "point of common coupling. " This point serves as the physical gateway for bidirectional power flow and as the boundary for ownership, responsibility, and power-quality supervision. Although physical access is decentralized through the core operating platform of the virtual power plant, these heterogeneous and dispersed resources are integrated into a unified, controllable, and schedulable virtual entity. At this level, the interaction between the VPP and external grid is manifested as a centralized, digitized, "virtual interface." Acting as a reliable and grid-friendly "virtual power source" or "virtual load," it participates in ancillary services such as peak shaving, frequency regulation, and demand response. The dynamic power input/output is expressed as follows:

$$P_{sch_to} = \begin{cases} P^t_{sch_toL}(\text{Injection into the load}) \\ P^t_{sch_toS}(\text{Absorbed from the grid}) \end{cases} \tag{12}$$

where *Pt sch_toL* denotes the scheduled power allocated to the downstream loads, and *Pt sch_toS* represents the scheduled power dispatched to coordinate with the upstream grid.

The absorbed or injected power of the grid scheduling center, together with the range of adjustable loads, must satisfy the following constraints:

$$\begin{cases} P^t_{sch_toL,\max} \leqslant P_{sch_to} \leqslant P^t_{sch_toS,\max} \\ \Delta P_{sch_to,\min} \leqslant \Delta P_{sch_to} \leqslant \Delta P_{sch_to,\max} \end{cases} \tag{13}$$

where ΔP_{sch_to} is the adjustable load.

2.2. VPP scheduling framework

In summary, the VPP is an efficient and reliable energy ecosystem constructed using digital means. The high heterogeneity of internal resources poses severe challenges for operational scheduling. According to the supply–demand characteristics of VPP resources across different periods, these heterogeneous resources are aggregated into two primary roles—the "supply side" and the "load side", which greatly reduces the complexity of the scheduling model.

Within this framework, DERs are abstracted into one virtual entity, mainly to reduce the dimensionality of the strategy space and enhance the scalability. This formulation implicitly assumes that supply side DERs constitute a cooperative coalition coordinated by the VPP operator, which is in line with common operational paradigms in current VPP implementations (Li et al., 2025). Moreover, in liberalized electricity markets where supply-side DERs may be owned by independent entities with competing interests, the model can be naturally extended to include internal competition. Specifically, individual DERs or clusters

on the supply side can be considered as multiple non-cooperative agents participating in an internal sub-game with the aggregated supply-side mixed strategy being the equilibrium outcome of this lower-level competition.

A bi-level RL framework was adopted, which divides the overall scheduling problem into two layers: “high-level decision-making” and “low-level optimization.” The constructed hierarchical reinforcement learning framework consists of an upper-level global coordination layer and a lower-level local execution layer. The upper level comprises a supply-side controller and a load-side controller. The former is responsible for formulating the overall electricity price and output allocation strategy in line with the system's economic objectives and supply–demand balance, whereas the latter plans the global response scheme based on price signals and load forecasts. Together, they constructed a price–load collaborative global scheduling strategy to guide the specific decisions of the lower level. The lower level consists of multiple subordinate units on the supply side and load side, which interact in real time through price signals and electric energy. The upper-level game captures the slower trends of strategic coordination under uncertainty, while the lower-level agents operate at a finer temporal granularity, continuously adjusting their actions in response to evolving system states. The overall framework is illustrated in Fig. 1.

3. Modeling of upper-level dynamic sequential game theory

Based on the division of roles into the supply side and the demand side, the supply-side pricing strategy directly influences the demand-side consumption behavior, whereas the demand-side response affects the supply-side revenue. Therefore, the supply side should play a leading role by participating in the VPP's optimization scheduling through a dynamic pricing mechanism. This mechanism guides the demand side to shift consumption patterns and alleviates grid congestion pressures. In response, the demand side dynamically adjusts its consumption schedule based on price signals to reduce costs and obtain demand response incentives, as shown in Fig. 2.

Based on the above analysis, this section develops a dynamic sequential game model with mixed strategies in which the supply and demand sides optimize their strategies through a two-stage sequential game. As the “leader,” the supply side takes a dynamic pricing approach to exert its dominant role, dividing the day into n pricing periods, each associated with a price level $\boldsymbol{\pi}=(\pi_1,\pi_2,..,\pi_n)$. The price strategy is represented by a probability distribution vector $\mathbf{p}=(p_1,p_2,..,p_n)$, where $p_i\in$ (Zhou et al., 2025) and $\sum_{i=1}^{N} p_i = 1$ hold. The demand side, acting as the “follower,” adjusts its consumption schedule based on the observed price strategy, aiming to minimize costs. This is expressed by strategy $\mathbf{q}=(q_1,q_2,..,q_n)$, with constraints $q_i\in$ (Zhou et al., 2025) and $\sum_{i=1}^{N} q_i = 1$.

Let the rated power of the supply side i be denoted as $Pi\ s$ and the rated power of the demand side be denoted as $Pi\ l$. An imbalance between the power generation and load sides of the grid threatens the safe and stable operation of the system, potentially leading to severe consequences such as frequency instability and voltage collapse. To address this, a power imbalance penalty factor, σ, is introduced in this study to quantify the loss cost associated with power imbalance between the

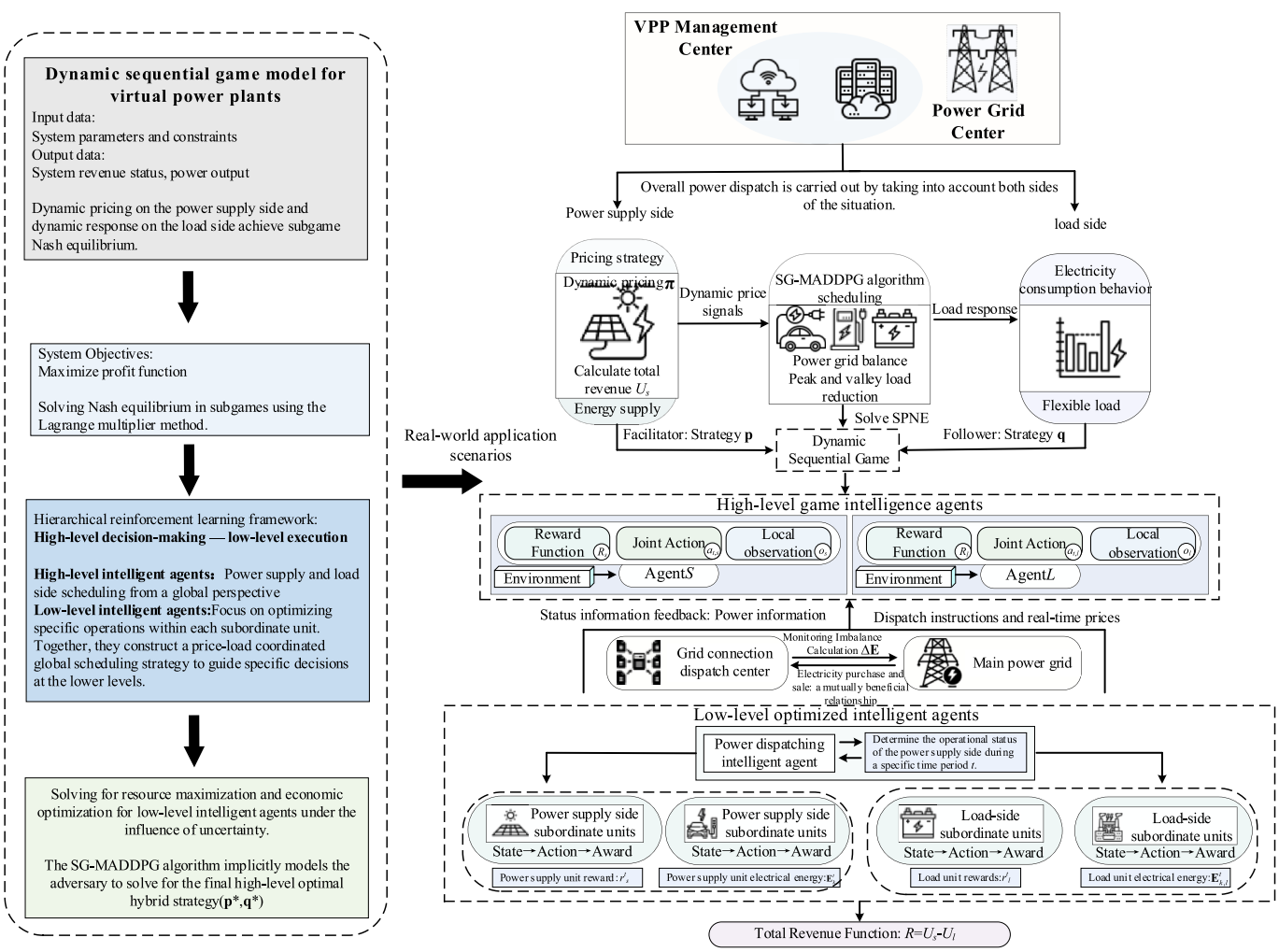

Fig. 1. Dynamic sequential game modeling for VPP and solving subgame Nash equilibrium using a hierarchical reinforcement learning framework.

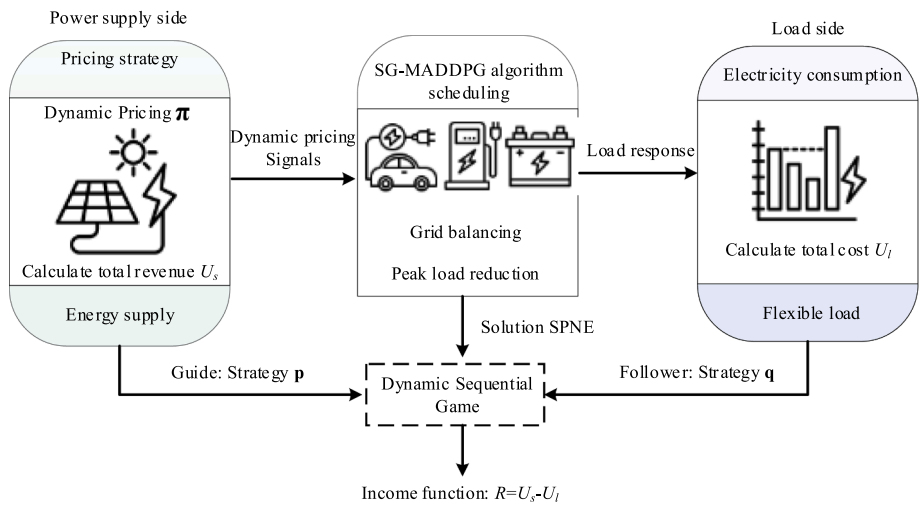

Fig. 2. System scheduling diagram.

generation and load sides. This is expressed as follows:

$$\sigma_s^i = \frac{\max\left(0, \sum_j P_s^j \cdot p_i - \sum_j P_l^j \cdot q_i\right)}{\sum_j P_s^j \cdot p_i}, \sigma_d^i = \frac{\max\left(0, \sum_j P_l^j q_i - \sum_j P_s^j \cdot p_i\right)}{\sum_j P_l^j \cdot q_i} \tag{14}$$

where $\sigma i\ s$ and $\sigma i\ l$ represent the unit penalty factors for the supply side i and demand side i, respectively. Based on the aforementioned penalty factors, the unit penalty cost C due to power imbalance is defined as a quadratic function of the factor σ, to maintain its convex increasing property, expressed as follows:$C_s^i = \lambda_1 \cdot \left(\sigma_s^i\right)^2, C_l^i = \lambda_2 \cdot \left(\sigma_l^i\right)^2$, λ is a game cost coefficient, and $\lambda \in$ (Zhou et al., 2025) to optimize total revenue (Huang et al., 2006). $Ci\ s$ and $Ci\ l$ represent the unit penalty costs for the power imbalance of the supply side i and demand side i, respectively. The convexity of this function ensures that the optimization problem has a unique global optimal solution.

Therefore, the mathematical expectation function can be designed as follows:

$$U_s = \sum_{i=1}^{N} p_i \left[\left(\pi_i - C_s^i\right) \cdot \min\left(p_i \sum_j P_s^j, q_i \sum_j P_l^j\right) \right], U_l = -\sum_{i=1}^{N} q_i \left[\left(\pi_i + C_l^i\right) \cdot \min\left(p_i \sum_j P_s^j, q_i \sum_j P_l^j\right) \right] \tag{15}$$

where U_s represents the mathematical expectation of the supply side's profit, and U_d represents the mathematical expectation of the demand side.

Considering the energy balance constraint, the expected supply or demand for the supply side or demand side, denoted as i, should not exceed the respective rated power and demand limits. Therefore, the following conditions must be satisfied:

$$T \sum_j P_s^i p_j \leqslant E_s^i, T \sum_j P_q^i q_j \leqslant E_l^i \tag{16}$$

where T is the time constant, and $Ei\ s$ and $Ei\ d$ represent the rated energies of the supply side i and load side i, respectively.

The optimization objective of the supply side is $\max_{\boldsymbol{p}} U_s(\boldsymbol{p}, \boldsymbol{q})$ and that of the demand side is $\min_{\boldsymbol{q}} U_l(\boldsymbol{p}, \boldsymbol{q})$. According to the game-theoretic framework, the first step is as follows: As the "follower," the demand side observes the supply side's mixed pricing strategy **p** and optimizes its consumption strategy **q*** to minimize costs:$\boldsymbol{q}^* = \operatorname{argmin} U_l(\boldsymbol{p}, \boldsymbol{q})$.

Second step: As the "leader," based on optimal strategy feedback from the demand side, the supply side solves for its mixed strategy p through backward induction to maximize its own profit.

3.1. Upper-level Nash equilibrium analysis

In the optimal scheduling of a VPP, the supply side acts as the leader by formulating a pricing strategy $\mathbf{p}=(p_1,p_2,..,p_n)$ corresponding to the probability distribution of electricity prices over n time intervals. As a follower, the demand side adjusts its charging strategy $\mathbf{q} = (q_1, q_2, \ldots, q_n)$ in response to observed price signals. Within the framework of the dynamic sequential game model, the profit function of the supply side $\boldsymbol{\pi}$ under electricity price i is defined as follows:

$$profit(\pi_i, p_i, q_i) = \left[\left(\pi_i - C_s^i\right) \cdot \min\left(p_i \sum_j P_s^j, q_i \sum_j P_l^j\right) \right] \tag{17}$$

Similarly, the cost function on the load side under electricity price $\boldsymbol{\pi}$ can be expressed as follows:

$$cost(\pi_i, p_i, q_i) = -\left[\left(\pi_i + C_l^i\right) \cdot \min\left(p_i \sum_j P_s^j, q_i \sum_j P_l^j\right) \right] \tag{18}$$

The Nash equilibrium requires that the mixed strategy **p** of the supply side maximizes its expected payoff, whereas the demand side cannot achieve a higher payoff by unilaterally deviating from **q**. Within a VPP, sub-units on both the supply and load sides are rational agents, each of which is interested mainly in its own decisions and in the collective behavior. Consequently, if agents on the same side are playing a game in a homogeneous strategy space, then the Nash equilibrium may be equivalently described by a post-equilibrium aggregate. This approach provides a possibility to model the system behavior with the help of a representative aggregate (Singh et al., 2025). In a mixed-strategy Nash equilibrium, the expected payoffs of all the pure strategies must be equal. Otherwise, the participants would adjust their probability distributions toward strategies that yield higher payoffs. Assuming that the demand side adopts a mixed strategy, $\mathbf{p}=(p_1,p_2,..,p_n)$, and the expected payoff of a pure strategy is denoted as $cost(\pi_i)$, the necessary and sufficient condition for the existence of a Nash equilibrium can be expressed as follows (see Appendix A for proof.):$cost(\pi_1) = cost(\pi_2) = \cdots = cost(\pi_n), \forall i$ and $q_i > 0$.

In summary, based on the Lagrange multiplier method and the definition of the Nash equilibrium, the necessary and sufficient condition for a mixed strategy $\mathbf{q}=(q_1,q_2,..,q_n)$ to constitute a Nash equilibrium is that the payoffs of all the pure strategies $q_i > 0$ are identical. In the context of VPP scheduling, this condition ensures equilibrium in the supply-side dynamic pricing strategy, guarantees the optimal response of the demand side, and contributes to optimizing grid load, economic

benefits, and traffic efficiency.

3.2. Solving the refined Nash equilibrium in upper-level subgames

Based on the aforementioned Nash equilibrium analysis, we adopt a two-stage solution approach to the mixed-strategy dynamic game model by employing the concept of a subgame perfect Nash equilibrium. The first step is to determine the optimal demand-side response in the second stage.

Given the probability distribution of electricity prices $\mathbf{q}=(q_1,q_2,..,q_n)$ determined by the supply side and the realized electricity price π_i associated with outcome p_i, the unit penalty cost for power imbalance on the demand side is denoted as $Ci\ d$. The total supply and demand are denoted by S and L, respectively. The demand-side optimization problem can be formulated as follows (for detailed solutions, see Appendix B.):

$$\min_{\boldsymbol{q}} \sum_{i=1}^{N} q_i \left[\left(\pi_i + C_i^i\right) \cdot \min(p_i \cdot S, q_i \cdot L) \right] \tag{19}$$

Similar to the second stage, p_i* forms a recursive functional dependency on itself that cannot be solved directly using algebraic methods. Therefore, reinforcement learning is a more reasonable solution.

3.3. MADDPG optimization and Improvement: SG-MADDPG

To derive the optimal mixed strategies ($\mathbf{p}$*,$\mathbf{q}$*) of the supply and demand sides within a VPP, this study relaxes the complete-information assumption of the traditional Stackelberg game. Under the constraints of stochastic electricity price distributions across time periods and dynamically coupled power flows, we developed an improved algorithmic framework called SG-MADDPG, which is built upon the MADDPG approach.

Random electricity price period distribution: $\tau \sim \mathscr{P}_\tau$ The power dynamic coupling satisfies: $\left|P_{\text{supply}} - P_{\text{load}}\right| \leqslant \varepsilon_t$.

The proposed model enables the distributed learning of subgame Nash equilibria under stochastic constraints. The key elements of the model are as follows:

Agent Design: Within the reinforcement learning framework, the supply and demand sides are modeled as agents that interact with the environment and exhibit both cooperative and competitive behaviors. These are denoted as the supply-side agent (Agent S) and demand-side agent (Agent L).

Action Space Design: To facilitate mixed-strategy learning, a gradient-based stochastic policy approach is adopted in which agents directly output the parameters of the probability distribution of their strategies. Specifically, the action of Agent S is defined as generating the probability distribution $\mathbf{p}$ of the electricity pricing strategy for the next time step, whereas that of Agent L is defined as generating the probability distribution $\mathbf{q}$ of the charging strategy for the next time step. Accordingly, the action space $\mathbf{A}$ is defined as follows:

$$a_{t,s} = \left[\boldsymbol{p}_t \in \mathbb{R}^n \middle| \sum_{i=1}^{N} p_i = 1, p_i \geqslant 0 \right], a_{t,l} = \left[\boldsymbol{q}_t \in \mathbb{R}^n \middle| \sum_{i=1}^{N} q_i = 1, q_i \geqslant 0 \right] \tag{20}$$

where $\mathbf{p}_t$ and $\mathbf{q}_t$ denote the strategies of Agent S and Agent L at time t, respectively. The agents interact with the environment by executing action strategies based on their outputs.

State-Space Design: According to the Markov property of a Markov game, the optimal decision of an agent in any state depends solely on its current state. The objective of the agent is to maximize its expected cumulative reward independent of the historical state information. Therefore, the key factors influencing the next-step decisions of Agent S and Agent L include: the current price information $\boldsymbol{\pi}$, the action $\mathbf{p}$ of the supply-side agent, and the action $\mathbf{q}$ of the demand-side agent. Accordingly, the state space $\mathbf{S}$ of the system at time t is defined as follows:$s_t = (\boldsymbol{p}_t, \boldsymbol{q}_t, \boldsymbol{\pi}_t)$.

In summary, the system state $\mathbf{s}_t$ is determined by the actions $\mathbf{p}_t$ and $\mathbf{q}_t$ of the agents together with the current price information π_t. The state transition function is implicitly defined by the environmental dynamics of the Markov decision process and is reflected through state updates; therefore, it does not require explicit modeling. This design effectively reduces the complexity of modeling state transitions and agent-environment interactions, thereby avoiding the need for explicit and cumbersome environmental modeling in reinforcement learning.

To guide both agents toward synchronous optimization, convergence to the Nash equilibrium, and the realization of their respective objectives, the reward function is constructed with two core components: (1) an objective-driven term that directly quantifies the profit maximization objective of Agent S or the cost minimization objective of Agent L and (2) a Nash equilibrium constraint term that ensures that the fundamental condition of a mixed-strategy Nash equilibrium is satisfied. The objective-driven reward serves as the primary performance goal, while the equilibrium constraint term is introduced as a regularization mechanism rather than a competing optimization objective. Under the equilibrium strategy, no agent can obtain a higher expected payoff or lower expected cost by unilaterally switching to any pure strategy. Accordingly, the reward function R_s of Agent S can be defined as follows:

$$R_s = \alpha U_s - \gamma \sum_{i=1}^{N-1} \left| profit(\pi_i) - profit(\pi_{i+1}) \right| \tag{21}$$

Agent L's reward function R_l is set as follows:

$$R_l = \mu U_l - \rho \sum_{i=1}^{N-1} \left| cost(\pi_i) - cost(\pi_{i+1}) \right| \tag{22}$$

Here, αU_s and μU_l denote the profit of the supply side and the expenditure of the demand side, respectively. The terms $\gamma \sum_{i=1}^{N-1} |profit(\pi_i) - profit(\pi_{i+1})|$ and $\rho \sum_{i=1}^{N-1} |cost(\pi_i) - cost(\pi_{i+1})|$ represent the Nash equilibrium guidance components for the two types of agents that drive the reward functions to converge rapidly toward the Nash equilibrium. The associated weight parameters were determined through sensitivity analysis experiments. Under these settings, the equilibrium constraint term and the objective-driven reward act complementarily. Specific parameter values are provided in Appendix F.

Optimization Objective Design: The optimization objective of each agent is to maximize the expected value of the discounted cumulative reward through policy optimization. Based on the principle of temporal-difference learning, the objective function L of Agent S is defined as the mean squared error loss of the Q function as follows:

$$\mathscr{L}_{i,s} = \mathbb{E}\ominus_{\boldsymbol{o},\boldsymbol{a},r_i,\boldsymbol{o}'} \left(R_i + \gamma Q_i\left(\mathbf{p}_t', \boldsymbol{\pi}_t', \boldsymbol{p}'\right) - Q_i(\boldsymbol{q}_t, \boldsymbol{\pi}_t, \boldsymbol{p}) \right)^2 \tag{23}$$

Agent L's objective optimization function L is defined as follows:

$$\mathscr{L}_{i,l}\ominus = \mathbb{E}\ominus_{\boldsymbol{o},\boldsymbol{a},r_i,\boldsymbol{o}'} \left(R_i + \gamma Q_i\left(\boldsymbol{q}_t', \boldsymbol{\pi}_t', \boldsymbol{q}'\right) - Q_i(\boldsymbol{p}_t, \boldsymbol{\pi}_t, \boldsymbol{p}) \right)^2 \tag{24}$$

Here, $Q_i(\mathbf{p}_t,\pi_t,\mathbf{p})$ and $Q_i(\mathbf{q}_t,\pi_t,\mathbf{q})$ represent the expected long-term returns of Agent S and Agent L, respectively, given the current electricity price and the strategies of all agents. $\gamma = 0.99$ denotes the discount factor, which balances the tradeoff between immediate and future rewards.

This paper proposes the SG-MADDPG algorithm, which improves the MADDPG framework while preserving its centralized training and decentralized execution paradigm. Critic networks have access to global information during training, whereas actor networks rely solely on local observations during execution. A dual-agent-independent actor–critic network architecture was designed to address the sequential nature of a two-stage supply–demand interaction.

1) Actor Networks: For the Actor network of Agent S, the input is its local observation space $\mathbf{o}_s$=($\mathbf{p}_t$,$\mathbf{q}_t$), and the output is the corresponding

action $a_s = \mu_s(\mathbf{o}_s;\theta_s)$. Similarly, for the Actor network of Agent L, the input is its local observation space $\mathbf{o}_l=(\mathbf{p}_t,\mathbf{q}_t,\pi t\ i)$, and the output is the corresponding action $a_l = \mu_l(\mathbf{o}_l;\theta_l)$.

2) Critic Networks: In virtual power plant scheduling, a sequential game relationship with mixed strategies exists between supply and demand. Traditional critic networks struggle to effectively handle the challenges arising from non-stationary environments and partial observability. To address this issue, we propose a GIADC that integrates graph neural network enhancement with adversarial dual-path optimization. By introducing a structured opponent-modeling mechanism, the network achieves an implicit representation of the opponent's strategies. Consider a multi-agent system composed of n agents under partial observability, where each agent i selects an action $at\ i$ based on its local observation $ot\ i$ and the environment state is denoted as $st\ i$. First, the system is modeled as a dynamic heterogeneous graph $g^t=(V,\varepsilon^t)$, where the nodes $\nu_i \in V$ represent the agents and the edges $et\ ij \in \varepsilon^t$ capture the interaction intensity between agents i and j at the current time step. Each agent maintains a policy network $\pi_{\theta i}(a_i|s_i)$ and a critic network $Q_{\varphi i}(\mathbf{o},\mathbf{a})$.

Then, to implicitly model the opponent's behavior, a dual-path encoding structure is incorporated into the critic, and a Graph Attention (GAT) mechanism is employed to aggregate neighborhood information. The feature representation of node ν_i is expressed as follows: $\boldsymbol{h}_i^{(0)} = MLP_{enc}\left(o_i^t \oplus a_i^{t-1}\right)$.

Edge weights are calculated using the query-key mechanism as follows:

$$\alpha_{ij} = \frac{\exp\left(\sigma\left(\boldsymbol{a}^{\mathrm{T}}\left[\boldsymbol{W}_q\boldsymbol{h}_j \| \boldsymbol{W}_k\boldsymbol{h}_j\right]\right)\right)}{\sum_{k\in\mathcal{N}(i)}\exp\left(\sigma\left(\boldsymbol{a}^{\mathrm{T}}\left[\boldsymbol{W}_q\boldsymbol{h}_j \| \boldsymbol{W}_k\boldsymbol{h}_k\right]\right)\right)} \tag{25}$$

where $\mathbf{a}$ denotes learnable parameters, $||$ denotes vector concatenation, and σ denotes the Leaky ReLU activation function.

Finally, distinct graph-reasoning paths were designed for the supply side and demand side. The supply-side strategy path is expressed as follows:$\boldsymbol{h}_s^{(m+1)} = GAT\left(\boldsymbol{h}_s^{(m)}, \left\{\boldsymbol{h}_l^{(m)} : \forall l \in \mathcal{N}(s)\right\}\right)$ The final embedding is $\mathbf{z}_s = MLP_s(\mathbf{h}M\ s)$. The critical output is expressed as follows:$Q_s(\boldsymbol{o}_s, \boldsymbol{a}_s) = f_\phi(\boldsymbol{z}_s \oplus \boldsymbol{p} \oplus \overline{\boldsymbol{q}_t})$ Here, $\overline{\boldsymbol{q}_t}$ denotes the mean action vector of load-side agents. The loader strategy path is expressed as follows:$\boldsymbol{h}_l^{(m+1)} = GAT\left(\boldsymbol{h}_l^{(m)}, \left\{\boldsymbol{h}_s^{(m)} : \forall s \in \mathcal{N}(l)\right\}\right)$ The final embedding was $\mathbf{z}_l = MLP_l(\mathbf{h}M\ l)$. The critical output is expressed as follows:$Q_l(\boldsymbol{o}_l, \boldsymbol{a}_l) = f_\phi(\boldsymbol{z}_l \oplus \boldsymbol{q} \oplus \overline{\boldsymbol{p}_t})$ Here, $\overline{\boldsymbol{p}_t}$ represents the weighted average of the pricing signals issued by the supply side agents.

The GIADC explicitly encodes inter-agent dependencies through a graph structure, thereby addressing the limitation of missing structural information in traditional critic networks. Moreover, the dynamic edge weight mechanism enables the model to adapt to non-stationarity induced by evolving strategies. In addition, because the supply side agent is the equilibrium result of an internal non-cooperative game between multiple DER owners, the GIADC can explicitly capture strategic interactions both within the supply side as well as across the different roles of agents.

3) Equilibrium Reinforcement Mechanism: Nash equilibrium constraint terms are incorporated into the reward functions of both Agent S and Agent L, guiding the profits and costs of the two types of agents toward equilibrium.

$$profit(\pi_i) = profit(\pi_j), cost(\pi_i) = cost(\pi_j) \tag{26}$$

This design ensures that the mixed-strategy updates satisfy the conditions for achieving a Nash equilibrium, thereby accelerating convergence and enabling the algorithm to reach the equilibrium more efficiently. The algorithm flow is presented in Appendix C.

4. Lower-level scheduling: MARL

4.1. Lower-level scheduling optimization objective analysis

In VPP power scheduling, the system must simultaneously satisfy three core constraints: power balance, temporal coupling, and resource competition. To this end, the complex scheduling problem was modeled as a hierarchical decision-making process. The output of the upper-level decision, manifested as the probabilistic scheduling strategy formulated by each unit in the day-ahead stage, serves as the basis and starting point for lower-level real-time scheduling. On the supply side, each unit divides a day into 24 time periods and maintains an output willingness distribution vector for each period based on the day-ahead electricity price forecast. This distribution satisfies the following probability constraints:$\sum_{k\in\mathscr{K}} p_{s,t}(k) = 1, \forall i \in \mathscr{G}, t \in \mathscr{T}$.

where $p_{i,t}(k)$ represents the probability that generation unit i selects output level k in time period t. As the 'price follower,' the load-side unit dynamically adjusts its electricity consumption plan in real time according to the observed supply-side price and output strategy, so as to optimize economy and flexibility. The electricity demand of load unit j in time period t can be expressed as $E_{i,t}=P_{i,rated}\bullet\Delta t_{i,t}$, where $P_{i,rated}$ is the rated power and $\Delta t_{j,t}$ is the actual operating duration. By adjusting $\Delta t_{i,t}$, flexible control of the electricity quantity can be achieved; the same applies to the supply-side unit.

Based on the above strategy, the upper-level scheduler counts the expected electric energy of all the units at operating time t:

$$E_s^t = \sum_{i\in\mathscr{G}} \mathbb{E}\left[E_{i,t}\right], E_l^t = \sum_{j\in\mathscr{L}} \mathbb{E}\left[E_{j,t}\right] \tag{27}$$

When a supply–demand imbalance occurs, the $Et\ s \neq Et\ l$, the grid-connected scheduling center will detect the power deviation $\Delta E^t = Et\ s - Et\ l$ and compensate for this difference through purchase/sale transactions with the main grid under economy and stability constraints, while feeding the compensation information back to the upper-level controller for strategy updating.

Ultimately, the power scheduling agent at the lower-level control center will, based on the actual balanced electric energy situation, specifically execute the power supply and consumption control for each distributed energy resource and load unit, thereby converting the upper-level probabilistic strategy into precise scheduling instructions that satisfy the actual supply–demand balance, completing the closed-loop process from collaborative decision-making to concrete execution, as shown in Fig. 3.

4.2. Constrained Markov game process analysis

To transform the VPP scheduling problem, which is characterized by high heterogeneity, multi-agent conflicts, and multi-source uncertainties, into a computationally tractable framework, the problem is formulated as a Constrained Partially Observable Markov Game (CPOMG). As such, the uncertainties within the system were precisely modeled to capture the stochastic nature of the real environment accurately.

4.2.1. Definition of a tuple in a Markov game

A Markov game can be described by the tuple $M=\langle\mathbf{s},\mathbf{a},\mathbf{p},\mathbf{r},\mathbf{c},\boldsymbol{\gamma},\boldsymbol{\Omega}\rangle$. Compared with deterministic models, an uncertainty space $\boldsymbol{\Omega}$ is introduced.

In this framework, the supply and load sides are abstracted into two types of agents to enable dynamic coordination of heterogeneous resources. The supply-side agent primarily represents renewable energy generation units, energy storage systems operating in discharge mode, and EV discharging to the grid. Its objective is to maximize the generation profit in response to upper-level price signals while handling uncertainties, such as the fluctuations of PV outputs and equipment failures. The load-side agent corresponds to industrial loads, energy

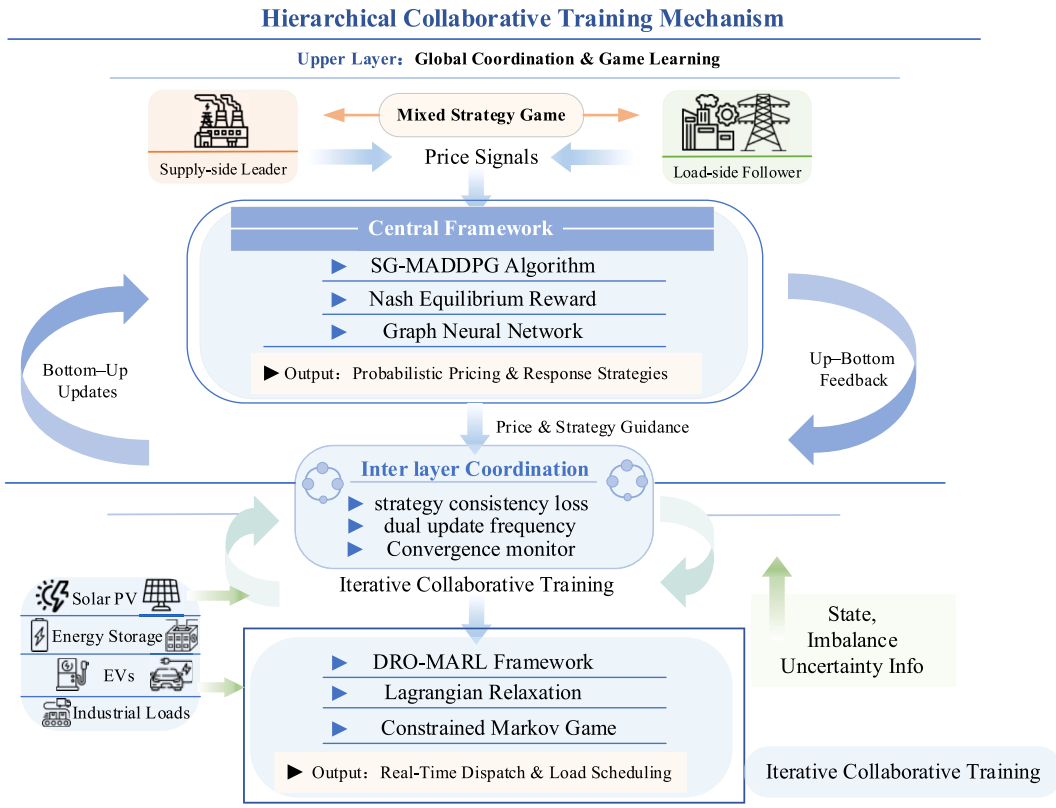

Fig. 3. Hierarchical collaborative training mechanism of the proposed bi-level MARL framework. The upper layer learns equilibrium-guided pricing and response strategies via a sequential game–based MARL model, while the lower layer performs robust real-time dispatch under uncertainty using distributionally robust MARL. Bidirectional information exchange enables coordinated learning and execution.

storage systems operating in the charging mode, and EVs in the charging mode. Its objective is to minimize electricity costs and dynamically balance supply and demand in response to electricity price disturbances. The two types of agents are non-cooperatively coupled through price signals and interact within the Markov decision process to achieve real-time market adaptation and global optimization.

s is the global state space. The state space reflects the system information that the scheduler can perceive in each period t. The state $s_t \in \mathbf{s}$ at time t includes: the current time, the status of each task, the current cumulative supply power and load power, the historical sequence of supply–demand deviations, and the feature variables characterizing the dynamics of uncertainty.

a is the action space of agent i. Each agent action was a binary decision.

$\mathbf{p}:\mathbf{s} \times \mathbf{a}^1 \times \cdots \times \mathbf{a}^N \times \mathbf{\Omega} \rightarrow \Delta(\mathbf{s})$ is the state transition function, which is now jointly determined by the joint actions of the agents and stochastic uncertainty.

r is the reward function of agent i.

c is the set of system constraints.

Ω is the uncertainty space, which encapsulates all exogenous and endogenous stochastic factors.

4.2.2. Multi-source uncertainty modeling

The uncertainties were quantified and categorized into two types to enable precise scheduling: exogenous and endogenous. Exogenous uncertainties are those that are caused by factors that are outside of the control of the internal system whereas endogenous uncertainties are those that are caused by internal events such as equipment faults. Given that the proposed framework is aimed at short-term operational scheduling under uncertainty, where decision robustness and computational feasibility are the priorities, this work takes a DRO perspective, assuming that the true distribution of the uncertain parameters is contained within an ambiguity set:

Exogenous uncertainty refers to stochastic factors beyond the control of the internal system.

The fluctuation in the renewable energy output (distributed photovoltaic generation) was modeled in the form of a day-ahead forecast plus random error:

$$P_{pv}^{t} = \widehat{P_{pv}^{h_t}} + \varepsilon_{pv}^{t}, \varepsilon_{pv}^{t} \sim \mathcal{N}\left(0, \sigma_{pv}^{2}\right) \tag{28}$$

where $\varepsilon t\ pv$ satisfies the time correlation:

$$\varepsilon_{pv}^{t+1} = \rho_{pv}\varepsilon_{pv}^{t} + \eta_t, \eta_t \sim \mathcal{N}\left(0, \sigma_{\eta}^{2}\right) \tag{29}$$

where ρ_{pv} is 0.8, and σ_{pv} is 0.05 $Pmax\ pv$.

The baseline load fluctuation (sudden increase/decrease in industrial load) is modeled in the form of a baseline + random disturbance + event shock:

$$P_{L,dem}^{t} = P_{L,dem}^{t,base} + \varepsilon_{L,dem}^{t} + J_t \tag{30}$$

where $\varepsilon t\ L_{,dem}$ also satisfies time correlation, and J_t represents rare shocks, which can be modeled as Poisson arrivals with parameter $J_t = B \bullet \mathbb{I}\{N_t > 0\}$.

Endogenous uncertainty refers to faults and other events originating from within the system.

Equipment failure or degradation is modeled as a random drift decline in the available power over time:

$$P_{avail}^{t+1} = P_{avail}^{t}(1 - \delta_t), \delta_t = \overline{\delta} + \varepsilon_t, \varepsilon_t \sim \mathcal{N}\left(0, \sigma_{\delta}^{2}\right) \tag{31}$$

4.2.3. Constraint handling

In an uncertain environment, satisfying hard constraints is more challenging. The constraints for the supply units and load units include

the following: load balance constraint: $\sum_{i=1}^{n} P_i x_{i,t} = D_t, \forall t$, where P_i is the rated power of each supply unit and D_t is the demand sequence of the load units. Continuous generation constraint: $T_{i,t} \leq T_{max,i}$. Start-stop frequency constraint: $N_{s,cum} \leq N_{max}$, where $N_{s,cum} = \sum_t \sum_i |x_{i,t} - u_{i,t-1}|$ and $u_{i,t-1}$ are the states of the previous time period.

To further enhance the flexibility and robustness of the lower-level RL model in handling hard constraints and exogenous and endogenous uncertainties, this study introduces the Lagrangian relaxation method. These constraints are relaxed into the objective function via Lagrangian multipliers λ_k, forming the following augmented Lagrangian function:

$$L(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) + \sum_k \lambda_k \nu_k(\boldsymbol{x}) \tag{32}$$

where $f(\mathbf{x})$ represents the original objective function and $v_k(\mathbf{x})$ quantifies the violation of the k-th constraint. By solving the dual problem $q(\lambda) = \max\lambda\min x L(\mathbf{x},\boldsymbol{\lambda})$, the relaxation method provides a lower-bound approximation to the original problem, and the optimal solution is approached when the multipliers λ are updated via sub-gradient ascent. The dual variable λ in the dual problem is updated using the sub-gradient method to ensure dual convergence: initialize $\lambda^0 \geq 0$. At each iteration k, it satisfies:

$$s^k = \nabla_\lambda q(\lambda^k) \approx g(x^*(\lambda^k)), \lambda^{k+1} = [\lambda^k + \alpha^k s^k]_+ \tag{33}$$

where $\alpha^k = c/\sqrt{k+1}$, and the sub-gradient of the dual function $q(\lambda)$ is bounded, $\|\nabla_\lambda q(\lambda)\| \leqslant \nu(x)$ holds for all $\lambda \geq 0$, so the average dual value $\overline{q}^K = (1/K)\sum_{k=1}^{K} q(\lambda^k)$ satisfies:

$$d^* - \overline{q}^k \leqslant \frac{\nu(x)^2 c + D^2 / c}{2\sqrt{K}} \tag{34}$$

where d^* is the optimal dual value, $D = \|\lambda^0 - \lambda^*\| \leqslant D_{\max}$. Strong duality ensures that the primal solution $x^* = \arg\min_a L(a, \lambda^*)$ is optimal and that the constraint violation converges to zero.

4.3. DROMARL (Distributed robust optimization Multi-Agent reinforcement learning core Algorithm)

4.3.1. Distributionally robust Critic design

The proposed distributionally robust Critic is designed to solve the following min–max problem based on the Wasserstein ambiguity set:

$$\theta_i^* = \underset{\theta_i}{\operatorname{argmin}} \sup_{\mathbb{P} \in \mathscr{B}_\varepsilon(\widehat{\mathbb{P}}_N)} \mathbb{E}_{\xi \sim \mathbb{P}} \left[\left(Q_{\theta_i}(s_i, a_i, a_{-i}) - y \right)^2 \right] \tag{35}$$

where $\mathscr{B}_\varepsilon(\widehat{\mathbb{P}}_N) \triangleq \{\mathbb{P} \in \mathscr{P} : W_1(\mathbb{P}, \widehat{\mathbb{P}}_N) \leqslant \varepsilon\}$ is the Wasserstein ball centered at the empirical distribution $\widehat{\mathbb{P}}_N$ with radius ε.

Theorem 1 **(Strong Duality and Equivalent Regularization).** Following the results of Mohajerin Esfahani et al. (2018) (Bertsimas et al., 2018; Mohajerin Esfahani and Kuhn, 2018), for the above Wasserstein DRO problem under mild regularity conditions, strong duality holds. The original problem is equivalent to the following regularized form: $\min_{\theta_i} \mathbb{E}_{\hat{\mathbb{P}}_N} \left[(Q_{\theta_i} - y)^2 \right] + \varepsilon \cdot \mathscr{R}(Q_{\theta_i})$, where $R(Q_{\theta i})$ is a regularization term related to the Lipschitz constant of the Q function.

Proof: According to Wasserstein's metric duality theorem, the supremum term can be transformed into a penalty term:

$$\sup_{\mathbb{P}: W_1(\mathbb{P}, \hat{\mathbb{P}}_N) \leqslant \varepsilon} \mathbb{E}_{\mathbb{P}}[f(\xi)] = \mathbb{E}_{\hat{\mathbb{P}}_N}[f(\xi)] + \varepsilon \cdot Lip(f) \tag{36}$$

where $Lip(f)$ is the Lipschitz constant of the function f. Under the present framework, $f(\xi) = (Q_{\theta i} - y)^2$. To ensure Lipschitz continuity, a gradient penalty mechanism was introduced.

Corollary (Equivalence of Gradient Penalty). Following Gulrajani et al. (2017) and Chen Y et al. (2023) (Gulrajani et al., 2017; Chen et al., 2023), by enforcing the Critic to satisfy 1-Lipschitz continuity, $\|\nabla_{\hat{s}} Q_{\theta_i}(\hat{s}, a)\|_2 \leqslant 1$, the original distributionally robust loss function can be approximated by the following more tractable gradient penalty form:

$$\mathscr{L}_{DR-Critic} = \mathbb{E}_{\hat{\mathbb{P}}_N} \left[(Q_{\theta_i} - y)^2 \right] + \lambda_{GP} \cdot \mathbb{E}_{\hat{s} \sim \mathbb{P}_{\hat{s}}} \left[\left(\|\nabla_{\hat{s}} Q_{\theta_i}(\hat{s}, a)\|_2 - 1 \right)^2 \right] \tag{37}$$

where $\hat{s}$ is a random interpolation point on the straight line between real samples and generated samples, $\hat{s} = \alpha s + (1-\alpha)s'$, with $\alpha \sim U$ (Zhou et al., 2025). This penalty term indirectly constrains the Wasserstein distance by constraining the gradient norm of the Critic, thereby achieving a balance between empirical risk minimization and the worst-case performance guarantee.

4.3.2. Low-variance target generation assisted by probabilistic world model

To further reduce the variance of the Q-value estimation under the worst-case distribution, a lightweight Probabilistic World Model (PWM) based on a Variational Recurrent Neural Network (VRNN) is introduced. This model is jointly trained end-to-end with the main Actor-Critic network. The PWM explicitly learns the conditional state transition distribution $p(s_{t+1}|s_t, \mathbf{a}_t, z_t)$, where z_t is a latent variable capturing the temporal correlations of uncertainty factors such as PV output, load fluctuations, and electricity price variations. The PWM is trained by maximizing the evidence lower bound, and its loss function is defined as:

$$\mathscr{L}_{PWM} = \mathbb{E}_{q(z|s,a)}[\log p(s'|z, a)] - D_{KL}(q(z|s, a) \| p(z)) \tag{38}$$

During training, the reparameterization trick is employed to ensure differentiable gradient propagation.

In the Critic update stage, the traditional single-step temporal difference target tends to produce high-variance estimates in heavy-tailed uncertainty environments. To address this issue, $H = 5$-step look ahead model rollout was performed using the trained PWM, generating $K = 10$ Monte Carlo sampled trajectories $\{\tau_k\}K$ *k-1* for each experience sample. Low-variance, informative samples are then selected via entropy regularization, focusing training on the most uncertain scenarios. Based on this procedure, the following low-variance robust target is constructed:

$$y_{ro} = r + \gamma \cdot \mathbb{E}_{k=1}^{K} \left[Q_{\theta_i} - \left(s_k', \pi(\mathbf{a}_k') \right) \right] + \beta \cdot \mathscr{H}(p(\cdot|s, a)) \tag{39}$$

where the last term is the policy-entropy regularization term, used to prevent the model from being overconfident in its predictions and to improve the policy's exploration stability. This mechanism seamlessly collaborates with the aforementioned Wasserstein DRO gradient penalty, and the multistep sampling trajectories provided by PWM effectively expand the coverage of samples in the experience replay buffer, making the constructed Wasserstein ambiguity set B_ε more distributionally representative, thereby significantly enhancing the stability of the Q-value estimation under the worst-case distribution.

4.3.3. Actor design based on uncertainty perception

A standard deterministic actor policy is prone to overconfident decision-making errors when faced with distributional shifts. This study proposes a two-stage actor design method to enhance a policy's perception of uncertainty.

Training stage: Parameterized noise $\varepsilon \sim N(0.\sigma^2 I)$ is injected after the input layer of the Actor network, with the noise intensity σ exponentially decaying with the training episodes, specifically $\sigma = \sigma_0 \bullet \rho^{episode}$, where $\rho = 0.999$. This operation is equivalent to Gaussian smoothing in the policy space, enabling the learned policy to possess inherent robustness to input perturbations, which theoretically corresponds to minimizing the following robust policy gradient objective:

$$\nabla J(\pi) \approx \mathbb{E}_\varepsilon[\nabla_\theta \log \pi_\theta(a|s + \varepsilon) \cdot Q(s + \varepsilon, a)] \tag{40}$$

Inference phase: Retain the Dropout layer used in the training phase (dropout probability $p = 0.1$) and perform $M = 8$ forward propagations. The final decision is obtained by averaging the output actions.

$$\hat{a} = \frac{1}{M}\sum_{m=1}^{M}\pi_\theta(s;\text{dropout}_m) \tag{41}$$

Simultaneously, the entropy $H(\pi_\theta(s))$ of the action distribution is computed as a quantitative indicator of uncertainty: when the entropy value exceeds a preset threshold τ, the system automatically switches to a conservative policy, thereby avoiding aggressive decisions in high-uncertainty scenarios such as sudden photovoltaic drops or electricity price jumps.

This design forms a closed-loop feedback with the overall DRO module. The uncertainty information perceived by the Actor is further fed back to the PWM to dynamically adjust the model sampling strategy, thereby continuously enhancing the overall robustness of the system under the worst-case distribution.

4.3.4. Robust Nash equilibrium guiding mechanism

In the stochastic game under consideration, the target strategy must satisfy Nash equilibrium conditions while remaining robust to distributional perturbations. Accordingly, a distributionally robust Nash equilibrium is defined.

Definition 1 **(Distributionally Robust Nash Equilibrium).** A mixed-strategy profile $\pi^*=(\pi^*_i,\pi^*_{-i})$ is called a distributionally robust Nash equilibrium if and only if, for all agents i and all possible distribution perturbations $P\in B_\varepsilon$, the following condition holds:

$$\mathbb{E}_{\mathbb{P}}\left[U_i\left(\pi_i^*,\pi_{-i}^*\right)\right]\geqslant\mathbb{E}_{\mathbb{P}}\left[U_i\left(\pi_i',\pi_{-i}^*\right)\right]\forall\pi_i'\in\prod_i \tag{42}$$

where U_i is the expected utility of agent i.

To achieve this distributionally robust Nash equilibrium, a robustness-guiding term is incorporated into the reward function.

$$R_{ro} = R_i + \eta\cdot\mathbb{E}_{\mathbb{P}\in\mathscr{B}_\varepsilon}\left[Var_j\left\{U_i\left(\pi_j\right)\right\}\right] \tag{43}$$

Proposition 1 **(Robustness Interpretation of Variance Penalty).** The variance penalty term $Var_i\{U_i(\pi_i)\}$ in the reward function is equivalent to introducing a risk-sensitive robustness term in policy optimization. Minimizing this variance term forces agents to select strategies whose payoffs exhibit smaller fluctuations across different distribution perturbations, thereby enhancing the strategies' stability.

Consider agent i's utility $U_i(\pi_i)$ under policy π_i as a random variable. Its variance can be decomposed as:

$$Var(U_i) = \mathbb{E}_{\mathbb{P}}\left[\left(U_i-\overline{U}_i\right)^2\right] \tag{44}$$

where $\overline{U}_i$ denotes the expected utility. By penalizing the variance, the mean–variance tradeoff objective is effectively optimized:

$$\max_\pi\left\{\mathbb{E}[U_i(\pi)] - \frac{\eta}{2}\cdot Var(U_i(\pi))\right\} \tag{45}$$

This is a typical risk-sensitive RL objective, whose solution is known to correspond to a robust optimal policy.

5. Simulation experiments and result analysis

To validate the effectiveness of the proposed bi-level MARL scheduling decision framework, this section conducts evaluation through numerical simulation experiments. The implementation of the experiments is done on the platform of the Matlab/Simulink in conjunction with the PyTorch library in Python to realize the MARL algorithm. The simulation scenario models a typical VPP system comprising distributed PVs, energy storage units, EV clusters, industrial loads, and a grid-connected scheduling center. The experiments incorporate real-world factors such as electricity price fluctuations, renewable energy output uncertainty, and load disturbances, and assess the performance of the framework in terms of global profit improvement, default risk reduction, and algorithmic convergence stability. All experiments are repeated 10 times, and the average values are reported to mitigate the influence of randomness. The comparison benchmarks include traditional centralized optimization approaches: model predictive control based on linear programming (MPC),conventional Stackelberg game method, standard MADDPG algorithm, and the deep end-to-end causal inference-based optimization scheduling method (DECI-VPP) (Geffner et al., 2202). Under unified parameter settings, five different scenarios are constructed, each adopting different baseline methods for comparative analysis, as summarized in Tab. 1.

The detailed experimental parameters and the calculation methods for key performance indicators (global profit, default risk, algorithm convergence speed, robustness, and equilibrium quality metrics) used in this paper are provided in Appendix F.

5.1. Experimental results and analysis

5.1.1. Robustness assessment in uncertain scenarios

Different levels of uncertainty are introduced (including PV output fluctuations of 5% to 20%, and load disturbances of 5% to 15%) to evaluate the performance of the proposed method across 100 Monte Carlo scenarios. The box plot of the corresponding global profit as the level of uncertainty increases is shown in Fig. 4.

As can be seen in the figure, when uncertainty increases, the profit boxes have shifted down (reduced profit) and the boxes have become wider (greater variability). Notably, the proposed method has the smallest downward shift, the narrowest box width and the fewest outliers under high uncertainty, and has much higher overall profit compared with the other methods. These results show the effectiveness of the DRO mechanism and GNN enhancements in improving the system robustness.

Table 2 below lists the Default Risk (DR) values for each method at different levels of uncertainty (5%, 10%, 15%, and 20%).

The DR analysis shows that as the level of uncertainty increases from 5% to 20%, the DR of all the methods shows a roughly linear upward trend. This is mainly because of the increased supply–demand

Table 1
Different scene settings and features.

Scenario	Method	Feature description	Applicable Scenarios
Scenario 1	MPC Method Based on Linear Programming	It supports linear inequality constraints on decision variables and state variables	Minimize operating costs after constraint processing
Scenario 2	Traditional Stackelberg game theory	Given a fixed leader power supply strategy **p**, solve for the optimal follower load side **q*** and substitute it into the solution for numerical optimization of **p***	Offline solution of SPNE benchmark
Scenario 3	Standard MADDPG algorithm	It adopts a centralized training and distributed execution framework, and uses a centralized critic to coordinate the strategies of each agent	Assessing the necessity of MARL collaboration
Scenario 4	DECI-VPP method	The solution is obtained by simultaneously identifying causal structures and estimating causal effects from observational data	Extracting causality from observation data
Scenario 5	This article's method	The upper layer uses the SG-MADDPG algorithm, the lower layer uses the DROMARL algorithm, and the hierarchical MARL sequential game theory is used to build a hierarchical optimization.	Nash equilibrium optimal scheduling

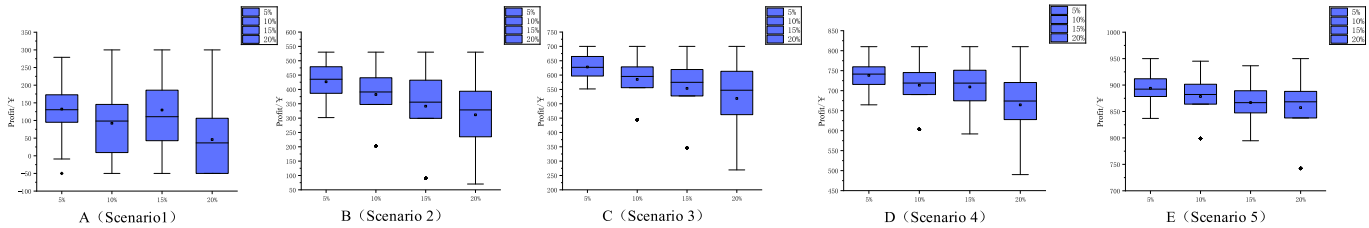

Fig. 4. Box plot of global profit under uncertainty level.

Table 2
Fault risk values in different scenarios.

Scenario /%	Uncertainty level 5%	Uncertainty level 10%	Uncertainty level 15%	Uncertainty level 20%
Scenario 1	6.0	9.0	12.0	15.0
Scenario 2	3.5	8.0	11.0	14.0
Scenario 3	3.0	5.5	7.5	9.5
Scenario 4	2.5	4.5	6.0	7.5
Scenario 5	2.0	2.8	3.4	4.2

imbalances (ΔP) caused by PV output fluctuations and load disturbances, which are further increased by equipment failures. The overall average DR results are as follows: The average DR of the proposed method is 3.1%, showing that the proposed method can reduce the average DR by 66.9% compared with the Stackelberg game method (9.1%), 52.3% compared with standard MADDPG (6.4%), and 40.3% compared with the DECI-VPP method (5.1%). These improvements are attributed to the DRO mechanism, which addresses the non-stationarity of the environment and policy oscillations, and the GNN-enhanced Critic network, which better models the inter-agent dependencies, which allows for rapid adjustments of electricity pricing strategies (**p**) and load responses (**q**).

5.1.2. Algorithm convergence analysis

To test the convergence performance of the proposed method, the algorithm is trained with fixed uncertainty levels, i.e. 10% fluctuation in PV output and 5% load disturbance. The cumulative reward curves for both supply and load sides are registered. Fig. 5 shows the convergence curves of the various methods.

The analysis shows that there is a cumulative increase in the rewards with the number of training episodes in all cases, although there is a significant variation in performance. The final reward values for Scenarios 1 through 4 increase successively, and are around −200, −100, −50, and close to 0, respectively. Scenario 5, which represents the proposed method, while achieving the final reward closest to zero, significantly outperforms the other scenarios, and also has the smoothest and most stable learning curve, with the fastest convergence and minimal oscillations. These results show a better learning efficiency and policy stability for the proposed approach.

5.1.3. Global profit and cost optimization comparison

Under the benchmark scenario without additional disturbances the global profits of the different methods are compared. Tab. 3 gives the average profit and cost of each method over a 24-hour period.

As can be seen in Fig. 6, the proposed method is consistently better than all the comparison approaches in terms of the important metrics. For the same amount of power supply, it greatly saves the cost on the load side, can achieve more accurate power supply–demand management, and finally realize profit on a global scale. The profit values of the comparison methods are 567¥, 611¥, 613¥, and 620¥, respectively, while the proposed method shows an increased profit of 711¥, which is a 14.7% increase compared to the best baseline and an increase of up to 25.3% compared to the worst performing method. These results reflect

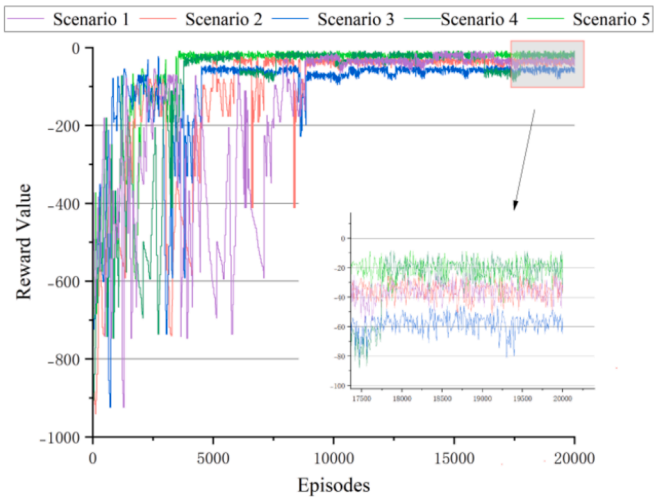

Fig. 5. The reward curves under the identical reward function setting reflect the training reward convergence of different methods. The differences in convergence levels demonstrate the performance gap in handling VPP uncertainties, satisfying system constraints, and achieving coordinated equilibrium behavior.

the obvious benefits of the proposed method in terms of overall performance and system economics, and establish the effectiveness and superiority of the proposed method in power scheduling and energy management applications.

5.1.4. Dynamic analysis of electricity prices and load response

Analysis of the dynamic response of the electricity prices and load curves on a typical day is presented. Fig. 7 shows the electricity price probability distribution together with the load adjustment curve derived with the proposed method.

The results show that the dynamic variation of the electricity price probability distribution is effective to represent the supply and demand situation, so that the user load curve can respond accordingly. Specifically, loads are reduced actively during high-price periods and moderately increased during low-price periods. This response mechanism greatly smooths the overall system load curve, reduces the peak-to-valley difference, reduces the pressure on grid operation, and greatly improves the stability and operation efficiency of the system.

5.1.5. Sensitivity analysis

A sensitivity analysis of key parameters is performed. Fig. 8 shows the differences in global profit and equilibrium quality measures in terms of the penalty coefficient λ.

Scenario 5 always has the largest global profit with different penalty coefficient λ, which reaches the maximum value of 976 when $\lambda = 0.5$, and has an improvement of 26.8%~34% compared with the other scenarios. Simultaneously, its metric of equilibrium quality is stably less than 0.01, which means that profit maximization in this scenario does not affect the stability of the system and reaches an optimal balance between economic benefits and operational performance. As the value

Table 3
Comparison of global profit and cost under different methods.

Method	MPC Method Based on Linear Programming	Traditional Stackelberg game theory	Standard MADDPG algorithm	DECI-VPP method	This article's method
Profits on the power supply side / ¥	1345	1354	1365	1372	1444
Load-side cost / ¥	664	640	643	645	635
Power imbalance penalty / ¥	114	103	109	107	98
Overall Profit / ¥	567	611	613	620	711

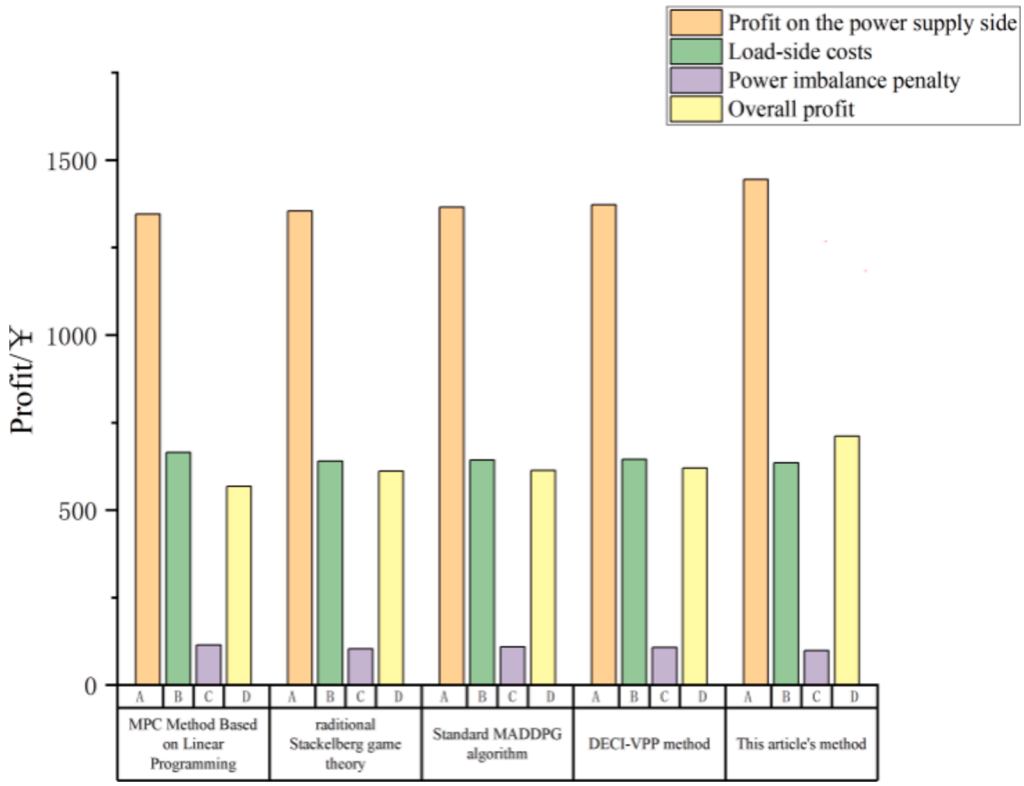

Fig. 6. Global Profits of Different Methods.

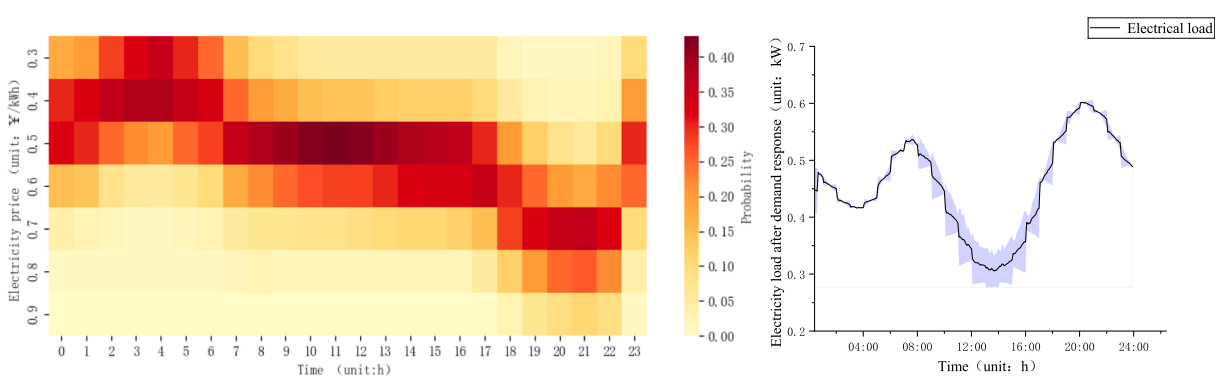

Fig. 7. Electricity price probability distribution heatmap and load response curve.

of λ varies from 0.1 to 1, the global profit curve of Scenario 5 shows the most gradual decrease, while the equilibrium quality measure is stable with little fluctuation. These results show that Scenario 5 is the least sensitive to parameter variations and has the most robust control strategy, and can maintain stable performance under different policy settings and market conditions.

5.1.6. Ablation experiment

To systematically determine the contribution of each key component in the proposed framework, ablation experiments are presented in this section. The performance of various variants is assessed under benchmark and uncertainty scenarios by selectively removing or modifying core modules of the framework. Experimental settings are based on those of Appendix E, with the same system parameters and uncertainty

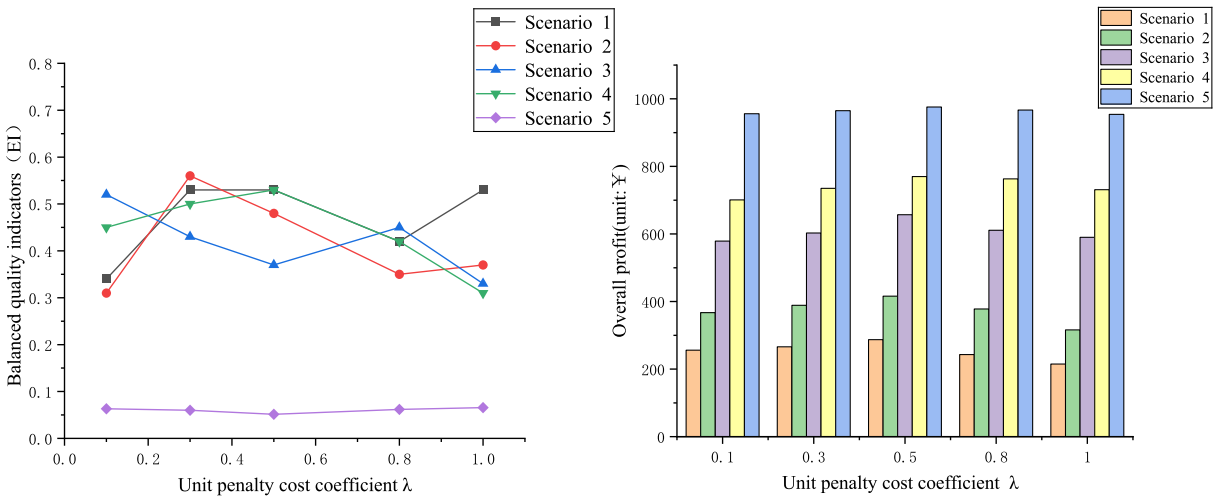

Fig. 8. Global profit and equilibrium quality indicators vary with the penalty coefficient λ.

models. The variants are as follows:

1) Proposed method- noGNN: eliminates the GNN-enhanced GIADC Critic, and replaces it with a standard fully connected Critic network.

2) Proposed method-MLP: encodes neighboring states using independent MLPs, without using any graph structure or message passing.

3) Proposed method-noNE Eliminates the Nash equilibrium constraint term from the reward function, leaving only the objective driven part.

4) Proposed method-noDRO: Eliminates the DRO component and relies only on the upper level game model for optimization.

5) Full proposed method – the entire framework, which is the baseline.

Each variant is trained for 20,000 iterations with fixed levels of uncertainty, with results averaged over 10 independent runs. Evaluation metrics include global profit, DR, number of iterations to convergence and deviation from the Nash equilibrium. Robustness is further evaluated using Monte Carlo simulations for 100 disturbance scenarios.

Based on the ablation results illustrated in Tab. 4, the critical role of each of the components in the proposed multi-agent reinforced learning double-layer scheduling framework can be verified in a systematic way. Under uncertain conditions, the full framework had a high global profit, and a DR of just 2.8%. In contrast, the performance of each variant significantly decreased: the DR of the noGNN variant increased to 4.5%, which is a 61% increase compared with the complete framework; the noNE variant further increased to 5.2%, which is an 86% increase. The noStr variant had the worst performance degradation with a DR of 6.1%, which is a 118% increase. Replacing the GNN module with an MLP led to a DR of 4.6%, which is 64% higher than the full framework. This stepwise increase in risk underlines the key role of the hierarchical multi-agent reinforced learning model in controlling environmental uncertainty. Specifically, the GNN module is effective in improving dependency modeling among intelligent agents, while replacing it with an MLP results in higher variance in profit, higher risk and slower convergence. Moreover, the Nash equilibrium constraint mechanism guarantees the stability of the system, and the performance deterioration of the system without the hierarchical strategy (noDRO) further proves the decisive effect of the structured decision-making framework on the coordination of complex multi-agent interactions.

5.2. Analysis of real-world cases

In order to further validate the practical applicability and effectiveness of the proposed bi-level MARL scheduling framework, an analysis based on a real-world pilot VPP project in China is presented in this section. The case was chosen from publicly available reports and industry data, and represents operational environments with high resource heterogeneity and with significant uncertainty. The analysis procedure includes data acquisition, model implementation, performance evaluation and comparative analysis. The proposed framework is applied to historical operational data to simulate optimized scheduling strategies and their performance is evaluated against actual operational records as well as benchmark methods in terms of overall profit improvement, reduction of DR, and enhancement of system stability. All analyses are based on publically available datasets for the 2023–24 period with the results averaged over 5 independent runs to reduce stochastic variability. Taking into account e.g. system scale, computational feasibility and data availability, the framework is implemented on a daily timescale. Benchmark comparisons are the traditional centralized scheduling approach used in real operations and the conventional Stackelberg game method. Detailed data sources and specific VPP parameters are given in Appendix F.

5.2.1. Case study of VPP in Suzhou industrial Park

The Suzhou Industrial Park VPP pilot focuses on industrial load response and renewable energy integration, which is confronted with large fluctuations in peak load and electricity price in the actual operation. The proposed method is applied for simulating a 24 h scheduling horizon. On a normal working day, the electricity consumption pattern in Jiangsu Province (dashed reference curve) shows two obvious peaks: a morning peak between 10:00 and 11:00 and an evening peak between 13:00 and 16:30. The park is equipped with 3660 public charging piles, 607 dedicated charging piles and 30 000 private charging piles, for a

Table 4
Performance comparison of different variants in ablation experiments.

Variant	Overall profit / ¥	Default risk /%	Number of convergence iterations	Nash Equilibrium Quality Indicators	Robustness
The method described in this article – noGNN	743	4.5	9000	0.25	57.8
Method used in this paper – MLP	727	4.6	8500	0.28	47.5
Method described in this article – noNE	720	5.2	7000	0.37	45.7
Method used in this article – noDRO	741	6.1	8000	0.12	35.9
Full text of this article's methods	776	2.8	4000	0.065	14.3

total installed capacity demand of 376 400 kVA. Considering a simultaneity coefficient of 0.6, the corresponding peak load is approximately 120 400 kW. Using this real-world data, an optimized scheduling is performed in the proposed framework, the specific results of which are summarized in Tab. 5.

The experimental results for the Suzhou case, as shown in Tab. 5, show the evident superiority of the proposed bi-level MARL scheduling framework in terms of multiple important metrics. In terms of economic performance, the global profit of 117.85 million CNY is obtained under the proposed method, which is 32.7% higher than that of the traditional Stackelberg game and 19.2% higher than standard MADDPG, indicating the superior ability of the proposed method in coordinated optimization. Regarding system stability, the power imbalance penalty is reduced to 39.96 million CNY, which is 6% lower than centralized MPC optimization, with a DR of 2.9%, which is significantly lower than 6.2% for traditional methods and 4.8% for baseline MADDPG. Notably, the method maximizes the supply-side profit and minimizes the load-side cost at the same time, which shows its capability of achieving a win–win situation for both suppliers and consumers. Compared with DECI-VPP, the full version of the proposed method further improves the global profit by 32.4%, which validates the crucial contribution of the value iteration module in long-term revenue optimization.

5.2.2. Case study of VPP pilot Zones in Anhui Province

The Anhui pilot zone emphasizes the involvement of EV clusters, where practical difficulties are uncertainty in the SOC of EVs and sudden variations in electricity prices. The proposed framework is used for simulation and optimization based on measured data from the Xuancheng VPP. Optimized scheduling is carried out to test the transferability and applicability of the strategy under conditions of multi-resource aggregation and interactions with the electricity markets. The results, which compare the output and power profiles of each unit in different scenarios, are shown in Fig. 9.

Fig. 9 shows the power output of each component included in the VPP in different scenarios during a typical day and their temporal variation trends. The results show that the system can perform effective scheduling and resource allocation while adapting to various operational requirements.

As shown in Table 6, the experimental results from the Anhui case also confirm the clear superiority of the proposed method for all the key performance metrics. Specifically, it enjoys a global profit of 24.78 millon CNY, which is 20.4% better than the second best method (DECI-VPP) and 28.5% better than the traditional Stackelberg game. The framework also achieves a true win–win result for both sides: supply side profit is maximized to 944.48 million CNY and load side cost is minimized to 853.09 million CNY. Regarding the system stability, the power imbalance penalty is reduced by 33.6% compared to standard MADDPG, and the quality metric, equilibrium, is close to the near-optimal level, which highlights the effectiveness of the proposed approach in both economic and operational dimensions.

5.3. Experimental conclusions

Experimental results show that the proposed bi-level MARL framework can significantly outperform benchmark methods such as centralized MPC, conventional Stackelberg game, standard MADDPG and DECI-VPP under uncertain market conditions. In benchmark scenarios, the framework significantly boosts the profit at the global level, showing significant improvements with respect to both MPC and Stackelberg game approaches. It also has faster convergence and significantly less deviations from the Nash equilibrium. Through parameter optimization, performance is further improved and the number of iterations needed for convergence is noticeably reduced. Ablation studies confirm that the DRO, GNN enhancement and Nash equilibrium constraint are critical to improving the robustness, modeling inter-agent interactions and guiding the equilibrium convergence, respectively. Under high uncertainty scenarios, the proposed framework exhibits a high robustness, indicated by narrower ranges of profit fluctuations and much lower DR than those of the Stackelberg game and the standard MADDPG approaches. Real-world case studies in China demonstrate significant profit improvements and a lot of risk reduction. Comparisons with international VPP projects also provide further validation of the generalisability of the framework, showing the clear growth in profits and effective control of risks. Overall, the proposed framework can effectively coordinate the heterogeneous resources, avoid the imbalance penalty, and support the low-carbon transition of power systems in accordance with the “dual-carbon” goals, which shows its good engineering application value. All the experiments were implemented with PyTorch 2.0, and the full code is publicly available at: https://github.com/IAMHUT/Hierarchical-Optimization-of-VPP-via-Sequential-Game-Based-Multi-Agent-Reinforcement-Learning.git.

6. Summary and Outlook

This work tackles the issues of heterogeneous resource scheduling and multi-stakeholder conflict coordination in VPPs under uncertain market conditions. It proposes a two-layer MARL scheduling framework based on a mixed strategy sequential game. Using a dynamic role reduction mechanism, the framework constructs a sequential game in a mixed electricity price strategy space, which achieves non-cooperative coupling between supply and demand sides. The framework uses a two-layer architecture: the upper layer optimizes mixed pricing strategies to dynamically balance conflicting interests and the lower layer formulates the scheduling problem as a constrained Markov game. Lagrangian relaxation is added to address power balance, temporal coupling and resource competition constraints and exogenous and endogenous uncertainties are explicitly quantified to ensure adaptability in real-time markets. Furthermore, the proposed SG-MADDPG algorithm incorporates GNN enhanced adversarial dual-path critic to address the environmental non-stationarity, DRO, curriculum learning and risk-sensitive reward shaping to improve training stability and operational robustness. Experimental results show that the online inference per time step consists of forward propagation through

Table 5
Performance Comparison of Different Methods in Suzhou Case Study.

Method	MPC Method Based on Linear Programming	Traditional Stackelberg game theory	Standard MADDPG algorithm	DECI-VPP method	This article's method
Profit on the power supply side / Million / ¥	242.42	247.34	265.65	261.02	276.05
Load-side costs / Million / ¥	113.54	114.63	121.34	125.03	118.24
Power imbalance penalty / Million / ¥	40.23	43.93	45.42	47.02	39.96
Overall profit / Million / ¥	88.65	88.78	98.89	88.97	117.85
Default risk /%	8.5	6.2	4.8	3.2	2.9
Balanced quality indicators	0.75	0.68	0.73	0.78	0.99

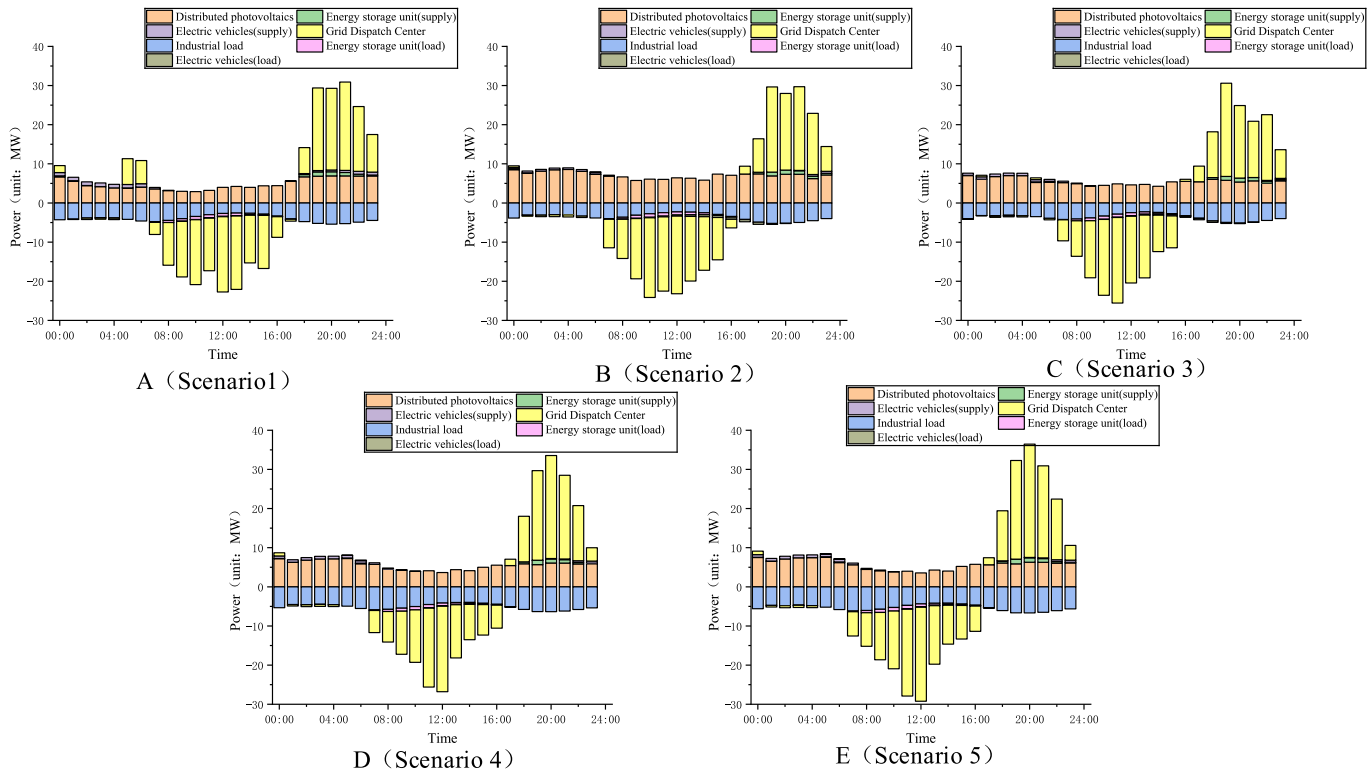

Fig. 9. Virtual power plant power dispatch status.

Table 6
Performance Comparison of Different Methods in Anhui Case Study.

Method	MPC Method Based on Linear Programming	Traditional Stackelberg game theory	Standard MADDPG algorithm	DECI-VPP method	This article's method
Profit on the power supply side / Million / ¥	937.38	940.94	933.02	955.83	944.48
Load-side costs / Million / ¥	859.18	859.32	840.41	860.93	853.09
Power imbalance penalty / Million / ¥	58.76	62.34	72.03	73.91	56.61
Overall profit / Million / ¥	19.44	19.28	20.58	20.99	24.78
Default risk /%	8.5	7.9	4.8	1.2	2.9
Balanced quality indicators	0.73	0.64	0.71	0.76	0.98

lightweight actor and critic networks, with an inference time of around 130–200 ms per time step on CPU-based industrial hardware, with GPU-based execution reducing further the latency to feasible scheduling intervals. The method has a significant impact on improving global profit and reducing DR by 65% under both baseline and high uncertainty scenarios. In a real-world VPP case study in China, it is shown to increase average profit by more than 20%, which supports the low-carbon transition of power systems.

Despite the benefits, there are some limitations to the framework. First, the computational complexity is high with high dimensionality uncertainty, which may be a challenge in real-time performance in large-scale deployments, especially in cases with tens of thousands of DERs. Second, the dual-agent simplification might result in suboptimal results in highly heterogeneous or strongly coupled multi-energy systems and does not fully capture dynamic real-world factors such as network latency or device failures. Additionally, agent interactions that involve data sharing, in particular, the transmission of price signals and state information, are potentially privacy risks. To overcome these limitations, future research efforts could focus on a number of improvements: (1) expanding the framework to multi-energy VPP systems that include thermal, hydrogen, and other forms of energy to increase applicability, (2) using distributed computing or model compression techniques to reduce the computational overhead and enable real-time deployment, (3) incorporating privacy-preserving technologies, such as blockchain, to secure data sharing, and (4) incorporating policy incentive designs and testing the generalizability with large-scale international datasets. These enhancements are expected to greatly enhance the practical deployability and application value of the framework.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

None

Appendix

A Proof of the necessary and sufficient conditions for Nash equilibrium:

The necessary and sufficient condition for a Nash equilibrium is:

$$cost(\pi_1) = cost(\pi_2) = \cdots = cost(\pi_n), \forall i \text{ and } q_i > 0 \tag{A1}$$

Necessity: Suppose $\mathbf{q}$=(q_1,q_2,..,q_n) is a Nash equilibrium and $q_i > 0$ holds. Under the Nash equilibrium, the expected cost of the demand side is minimized and a higher payoff cannot be obtained by adjusting $\mathbf{q}$. Considering the optimization problem of the demand side:

$$\min_{\boldsymbol{q}} U_l = -\sum_{i=1}^{N} q_i \cdot cost(\pi_i) \tag{A2}$$

Using the Lagrange multiplier method, construct the Lagrange function as follows:

$$L(\boldsymbol{q}, \lambda) = \sum_{i=1}^{N} q_i \cdot cost(\pi_i) + \lambda\left(1 - \sum_{i=1}^{n} q_i\right) \tag{A3}$$

Take the partial derivative for each q_i as follows:

$$\frac{\partial L}{\partial q_i} = cost(\pi_i) - \lambda = 0, \forall i \text{ and } q_i > 0 \tag{A4}$$

Therefore, $cost(\pi_i) = \lambda, \forall i$ and $q_i > 0$.

This implies that, for all pure strategies π_i of $q_i > 0$, their expected payoffs $cost(\pi_i)$ must be identical and equal to a constant λ. Otherwise, if there exist strategies i and j such that $cost(\pi_i) > cost(\pi_j)$ holds, the supply side could increase q_i and decrease q_j to improve the overall payoff U_s. This contradicts the definition of the Nash equilibrium, which precludes unilateral improvement; therefore, the claim is established.

Sufficiency: Assumption

$$cost(\pi_1) = cost(\pi_2) = \cdots = cost(\pi_n) = C \tag{A5}$$

where c is a constant and $\mathbf{q}$=(q_1,q_2,..,q_n) satisfy $\sum_{i=1}^{N} q_i = 1$ and $q_i > 0$.

Simultaneously, the cost on the load side is as follows:

$$U_l = \sum_{i=1}^{N} q_i cost(\pi_i) = \sum_{i=1}^{N} q_i \cdot c = c \cdot \sum_{i=1}^{N} q_i = c \tag{A6}$$

Regardless of how $\mathbf{q}$ is adjusted, the total cost, $U = c$, remains unchanged. Therefore, the supply side cannot increase its payoff by altering $\mathbf{q}$. This result verifies that the definition of the Nash equilibrium is satisfied.

B Nash equilibrium solution Process

Solving for Nash Equilibrium:

$$\min_{\boldsymbol{q}} \sum_{i=1}^{N} q_i \left[\left(\pi_i + C_l^i\right) \cdot \min(p_i \cdot S, q_i \cdot L) \right] \tag{B1}$$

Assuming that the electricity price π_i is known, which implies that the second stage constitutes a complete information setting, the demand side optimizes q_i with the objective of minimizing its cost. At this stage, the Lagrangian multiplier method is employed to construct the corresponding Lagrangian function as follows:

$$L(\boldsymbol{q}, \lambda) = \sum_{i=1}^{N} q_i \left[\left(\pi_i + C_l^i\right) \cdot \min(p_i S, q_i L) \right] + \lambda\left(1 - \sum_{i=1}^{n} q_i\right) \tag{B2}$$

In the Lagrangian function, partial derivatives are taken with respect to each q_i and set to zero. When the function involves $p_i S > q_i L$, the expression can be written as follows:

$$L(\boldsymbol{q}, \lambda) = \sum_{i=1}^{N} q_i \left[\left(\pi_i + \lambda_2 \left(\frac{q_i L - p_i S}{q_i L}\right)^2\right) \cdot p_i S \right] + \lambda\left(1 - \sum_{i=1}^{n} q_i\right) \tag{B3}$$

The partial derivative result is expressed as follows:

$$p_i S \cdot (\pi_i + \lambda_2) - \lambda_2 \cdot \frac{(p_i S)^3}{L^2 q_i^2} - \lambda = 0 \tag{B4}$$

Solving for q_i*:

$$q_i{}^* = \left(\frac{\lambda_2 (p_i S)^3}{L^2 [p_i S(\pi_i + \lambda_2) - \lambda]}\right)^{\frac{1}{2}} \tag{B5}$$

when $p_i S < q_i L$ is in the function, the function expression is as follows:

$$L(\boldsymbol{q},\lambda)=\sum_{i=1}^{N}q_i\left[\left(\pi_i+\lambda_2\left(\frac{q_iL-p_iS}{q_iL}\right)^2\right)\cdot q_iL\right]+\lambda\left(1-\sum_{i=1}^{n}q_i\right) \tag{B6}$$

The partial derivative result is expressed as follows:

$$2L(\pi_i+\lambda_2)q_i-2L\lambda_2\left(\frac{p_iS}{L}\right)-\lambda \tag{B7}$$

Solving for q_i*:

$$q_i{}^*=\frac{2\lambda_2p_iS+\lambda}{2L(\pi_i+\lambda_2)} \tag{B8}$$

The analysis indicates that q_i* is not only functionally dependent on the supply-side strategy p_i, but also on the cost λ of the demand side at the Nash equilibrium under pure strategies. However, λ exhibits functional dependence on q_i* as follows:

$$\lambda=cost(\boldsymbol{p}^*,\boldsymbol{\pi}) \tag{B9}$$

Therefore, q_i* inherently forms recursive relationships. This circular dependency structure makes it impossible to directly derive q_i* from closed-form analytical expressions. Therefore, this study employs a deep reinforcement learning approach in which equilibrium strategies are obtained through progressive numerical approximation until convergence is achieved.

By substituting the optimal solution q_i* as a condition in the analytical expression of the first stage, we can derive the optimal strategy for the supply side in the first stage. Given the response function $\mathbf{q}^*(\mathbf{p})$ of the demand side, the supply side optimizes the probability distribution $\mathbf{p}$ of electricity prices, with the unit penalty cost of the supply–demand imbalance denoted as *Ci s*. Thus, the supply side optimization problem can be formulated as follows:

$$\max_{\boldsymbol{p}}\sum_{i=1}^{N}p_i\left[\left(\pi_i-C_s^i\right)\cdot\min(p_iS,q_iL)\right],\ \textbf{(B10)}$$

By substituting the derived expression of q_i into the penalty term, the supply side optimizes p_i to maximize profit. By taking the partial derivative of p_i and solving it under the given constraints using the Lagrange multiplier method, an analytical expression for optimal p_i* can be obtained.

C detailed training procedure of the SG-MADDPG Algorithm

1) Initialize the actor and critic networks and target network parameters $\{\theta_s, \theta_l, \theta_s', \theta_s'\}$.

2) In each iteration, Agent S executes action $\mathbf{p}_t$, after which the environment generates real-time demand. Subsequently, Agent L performs action $\mathbf{q}_t$ and stores the experience $(s,\mathbf{a},r_1,\ldots,r_N,s')$ in a shared replay buffer D.

3) Dynamic Graph Construction: At each step, the node features $\mathbf{h}m+1\ s$ and $\mathbf{h}m+1\ l$ and the edge weights $\alpha i\ j$ are updated based on the current observations. These updated representations are then processed through L layers of GAT to generate node embeddings, followed by graph convolutional encoding.

4) A mini-batch is sampled from D for the parameter update. The critic network is optimized by minimizing the temporal-difference (TD) error L_i, where the target value is computed as follows:

$$\begin{cases} y_s=R_i+\gamma Q_i\left(\mathbf{p}_t',\boldsymbol{\pi}_t',\boldsymbol{p}'\right)\\ y_l=R_i+\gamma Q_i\left(\boldsymbol{q}_t',\boldsymbol{\pi}_t',\boldsymbol{q}'\right)\end{cases} \tag{C1}$$

The strategy gradient for Actor network updates is expressed as follows:

$$\nabla_{\theta_i}J_i\approx\mathbb{E}_{\boldsymbol{o}}\left[\nabla_{\theta_i\mu\theta_i}(o_i)\cdot\nabla_{a_i}Q_i^{\mu}(\boldsymbol{o},\boldsymbol{a}_1,\ldots,\boldsymbol{a}_i,\ldots,\boldsymbol{a}_N)\big|_{\boldsymbol{a}i=\mu\theta_i(o_i)}\right] \tag{C2}$$

Final target network soft update is expressed as follows: $\theta'\leftarrow\tau\theta+(1-\tau)\ \theta'$.

The Fig. C1 below shows the framework of the SG-MADDPG algorithm:

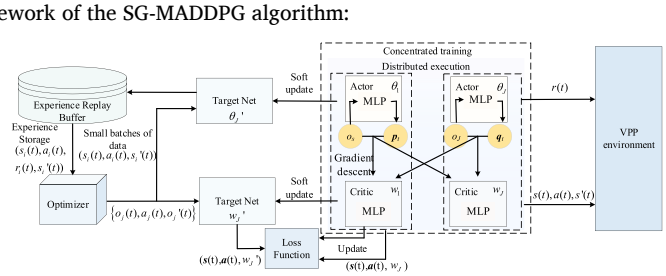

Fig. C1. Algorithm framework diagram

D Fixed-Point convergence and regret bounds of SG-MADDPG in Mixed-Strategy Markov Games

This section proves that, under certain mild assumptions, the proposed algorithm converges to the fixed point corresponding to a subgame-perfect Nash equilibrium and provides an average regret bound for the algorithm. The analysis builds on the theory of MARL in sequential games and extends the standard MADDPG framework to handle heterogeneous agents, non-stationary environments, and uncertainties in VPPs.

Under mixed strategies, the policy π_i:$S\rightarrow\Delta(A_i)$ of each agent i maps states to probability distributions over actions. The joint policy is denoted by $\pi=(\pi_1,\bullet\bullet\bullet, \pi_N)$. The value function of agent i is defined as:

$$V_i^{\pi}(s) = \mathbb{E}\left[\sum_{t=0}^{\infty}\gamma^t r_i(s_t,a_t)|s_0 = s,\pi\right] \tag{D1}$$

The Q function is:

$$Q_i^{\pi}(s,a) = \mathbb{E}\left[r_i(s,a) + \gamma V_i^{\pi}(s')|\pi\right] \tag{D2}$$

Assumption 1 **(Boundedness and Lipschitz Continuity):** The reward function r_i is bounded, i.e., $|r_i(s,a)|\leq R_{\max}$. The transition probability P and the value function $V\pi$ i are Lipschitz continuous with respect to the policy π:$\left\|V_i^{\pi} - V_i^{\pi'}\right\|\leqslant L_V\|\pi-\pi'\|$.

Here $L_V > 0$ is the Lipschitz constant. The state and action spaces are finite.

Assumption 2 **(DRO Robustness):**The DRO optimizes over the worst-case distribution within the Wasserstein ambiguity set P, ensuring that the sensitivity of the value function to uncertainty is bounded: $\sup_{\rho\in\mathscr{P}}\left|V_i^{\pi}(s;\rho) - V_i^{\pi}(s;\rho_0)\right|\leqslant\varepsilon_{DRO}$.

Here ρ_0 represents the nominal distribution.

These assumptions are reasonable in the VPP scenario: the state space corresponds to power balance and price signals, the action space corresponds to scheduling decisions, and the uncertainty is quantified through DRO.

0D.1 Fixed-Point Convergence:

Convergence of the SG-MADDPG algorithm to the fixed point of a SPNE under mixed strategies is theoretically proven. An SPNE is defined as a joint policy profile π^* that constitutes a Nash equilibrium $\forall i,s,\pi_i' : V_i^{\pi^*}(s)\geqslant V_i^{\left(\pi_i'(\pi_{-i}^*)\right)}(s)$ in every subgame, with the sequential structure ensuring backward-induction compatibility.

Theorem D.1 (Fixed-Point Convergence): Under Assumptions 1 and 2, SG-MADDPG converges with probability 1 to the fixed point π^* of a SPNE, i.e., there exists an iteration step K such that for all $k > K\|\pi^{(k)}$- $\pi^*\|,\|\pi^{(k)}$- $\pi^*\|\leq\varepsilon$, where $\varepsilon > 0$ can be arbitrarily small. The convergence rate is $O(1/\sqrt{k})$.

Proof: The lower-level MARL converges to a fixed point.

The lower level is a constrained Markov game, using Lagrange multipliers λ to handle constraints such as power balance: extending the reward $r'i = ri$-λg_i, where $g_i > 0$ is the constraint violation. MADDPG Update: Actor

$$\mu_i^{(k+1)} = \mu_i^{(k)} + \alpha\nabla_{\theta_i}J(\theta_i) \tag{D3}$$

Critic uses least squares fitting:

$$Q_i^{(k+1)} = \mathrm{argmin}_Q\mathbb{E}\left[(r_i + \gamma Q_i(s',a') - Q(s,a))^2\right] \tag{D4}$$

GIADC introduces a dual-path approach: the positive path optimizes the local agent, while the negative path simulates the adversary, decoupling non-stationarity.

$$\left\|Q_i^{(k)} - Q_i^*\right\|\leqslant(1-\beta)^k\left\|Q_i^{(0)} - Q_i^*\right\| + O(L_Q/\beta) \tag{D5}$$

Where β is the contraction rate. By assumption 2, the gradient variance is bounded, leading to policy gradient convergence:$\mathbb{E}\left[\left\|\nabla J^{(k)} - \nabla J^*\right\|^2\right]\rightarrow 0$. Combining DRO, the overall convergence is to π^* at a rate $O(1/\sqrt{k})$.

0D.2 Regret Boundary:

The regret measure is the cumulative difference between the algorithm and the optimal strategy. In mixed-strategy Markov games, the T-step regret of agent i is:

$$\mathrm{Reg}_i^T = \max_{\pi_i'}\sum_{t=1}^{T}\left(V_i^{\left(\pi_i',\pi_{-i}^{(t)}\right)}(s_t) - V_i^{\pi(t)}(s_t)\right) \tag{D6}$$

Theorem D.2 (Average Regret Bound): Under Assumptions 1–2, the average regret of SG-MADDPG satisfies:

$$\mathbb{E}\left[\mathrm{Reg}_i^T/T\right]\leqslant O\left(\sqrt{|\mathscr{S}||\mathscr{A}|(1-\gamma)^{-1}\log T/T} + \varepsilon_{DRO}\right) \tag{D7}$$

This is a sublinear regret.

Proof: Using online gradient descent analysis. Actor update is equivalent to no-regret learning: the policy space $\prod_i = \Delta(\mathscr{A}_i)^{|\mathscr{S}|}$ is convex. Gradient estimate $\widehat{\nabla}J^{(t)} = \nabla J^{(t)} + \xi_t$, where noise ξ_t and bounded variance σ^2. By online convex optimization, the regret bound is $O\left(\sqrt{T\log\left|\prod_i\right|}\right)$. In Markov games, this is extended to the discounted horizon: effective horizon H=$O(1/1$-$\gamma)$, total regret $\mathrm{Reg}^T\leqslant O\left(\sqrt{HT|\mathscr{S}||\mathscr{A}|\log T}\right)$. DRO adds an additive term of $\varepsilon_{DRO}T$. The sequential structure is propagated through backward induction: the regret bound of the upper level is passed to the lower level by multiplying by the Lipschitz constant L_V. After averaging, the bound is obtained.

To demonstrate significant improvement over the traditional Stackelberg game formulation, this paper analyzes the performance bounds of SG-MADDPG in repeated sequential games under uncertainty. Under standard assumptions commonly adopted in MARL regret analysis. Traditional Stackelberg models, which assume pure strategies and perfect information, may incur unbounded regret in repeated, uncertain environments due to a lack of dynamic adaptation. In contrast, the proposed framework, by incorporating mixed strategies and MARL, achieves no-regret learning with a sub-linear bound.

Theorem D.3 (Regret Bound in Repeated Game): For the repeated game over T time horizons, under assumptions 1 and 2, the average regret of an agent satisfies:

$$\mathrm{Regret}_T^i = \max_{\pi^i} \sum_{t=1}^{T} \left(V_t^{\pi^i,\pi^{-i}}(s) - V_t^{\pi_t}(s) \leqslant O\left(\sqrt{T\log T} + \varepsilon_{DRO} \right) \right) \tag{D8}$$

Proof: Extending Theorem D.2 to the repeated setting, the analysis employs online gradient descent. The actor updates are no-regret in the graphical policy space, and the GIADC provides β-contraction to decouple non-stationarity. By aggregating the regret over T rounds via the Hoeffding inequality, a high-probability bound is obtained. The DRO supremum ensures robustness against uncertainty, yielding sub-linear average regret.

From this regret bound, it can be inferred that the proposed method achieves asymptotic efficiency in the repeated VPP game: as $T \to \infty$, the average regret approaches zero, implying convergence to a robust subgame perfect Nash equilibrium despite uncertainties. Traditional Stackelberg equilibria often incur $O(T)$ regret in repeated uncertain games due to their static pure strategies. The bound for the proposed method guarantees asymptotic optimality, which correlates with the empirical profit increase and risk reduction observed in VPP scheduling.

E robust Reward-Guided local convergence of DROMARL

Theorem E1: Under standard assumptions (such as the concavity of the payoff function and the compactness of the policy space), the DROMARL algorithm guided by the robust reward described above can locally converge to a partially robust Nash equilibrium.

First, it is proven that, with a fixed opponent policy π_{-i}, the individual Bruker optimization problem for each intelligence i is a strongly convex problem, and its solution exists and is unique. Second, it is proven that the iterative update process of the entire policy profile constitutes a quasi-monotonic variational inequality. Using Rosen's (1965) classic conclusion on the existence of equilibrium points in concave games, it can be proven that the algorithm can converge to an equilibrium point.

F experimental parameter Settings

System parameter configuration:

The VPP system comprises 5 distributed photovoltaic units, 3 energy storage units, 20 electric vehicles, 2 industrial load units, and 1 grid-connected dispatch center. The simulation lasts 24 h, divided into 24 time periods, each lasting 1 h. Electricity prices are simulated based on real electricity market data, with a base price of 0.5 yuan/kWh and a fluctuation range of [0.3, 0.8] yuan/kWh, exhibiting a random walk model due to market uncertainties, as shown in Table F1 below.

Table F1
Virtual power plant system parameters.

Data types	Value	Unit
Distributed photovoltaic units	200	kW
Energy storage unit	150	kWh
Initial value of SOC	50%	/
Electric vehicles	500	kWh
Initial SOC distribution range	[20%,80%]	/
Industrial load unit	150	kW
Load adjustment range	[-30%,30%]	/
Power exchange at grid-connected dispatch center	[-100,100]	kW
Basic electricity price	0.5	￥/kWh
Electricity price fluctuation range	[0.3,0.8]	￥/kWh

Uncertainty factors are modeled as follows:

1) Renewable energy output uncertainty: Distributed photovoltaic output is based on historical meteorological data, with Gaussian noise added (mean 0, variance 10% of predicted value), and time correlation is considered, with coefficient ρ_{pv} of 0.8.

2) Load disturbance: Industrial load baseline is from the IEEE 33-node distribution network dataset, with random disturbances added based on a Poisson process, an average event interval of 6 h, and an impact amplitude of ± 20 kW.

3) Market electricity price uncertainty: Based on the random walk model of Equation 54, regression coefficient $\varphi = 0.05$, fluctuation term $\sigma_\pi = 0.02$, and jump probability 0.1.

4) Internal uncertainty: Equipment failure probability 0.05/period, power degradation rate 2%/failure.

In reinforcement learning training, considering that hyperparameters significantly affect training stability, convergence speed, and final performance, hyperparameters directly related to the learning process are given. At the same time, the setting of the coefficients of the reward function is crucial; inappropriate coefficients can lead to an imbalance in the influence of certain reward terms on the agent's decision-making. Specifically, the Actor network adopts a multilayer perceptron (MLP) architecture, which consists of three fully connected layers: the input layer dimension matches the observation space size; the first hidden layer contains 256 neurons; and the second hidden layer contains 128 neurons, both employing ReLU activation. The output layer dimension corresponds to the action space size, with a Softmax function applied to enforce probability distribution constraints. The total number of parameters is approximately 15,000.

In the GIADC architecture, the Critic network for each agent is based on a GAT comprising $L = 2$ GAT layers. Each GAT layer utilizes multi-head attention, where the query-key vector dimension per head is 64. The node features are initialized with the observation dimension and aggregated via GAT into a hidden representation h^m. Subsequently, an MLP generates the final embedding for either the supply-side or load-side. The Critic outputs a scalar Q-value through a linear layer. The total parameter count for the entire GIADC ranges approximately from 50,000 to 60,000.

The time complexity of the proposed method is primarily dominated by the GAT operations and Critic updates. For N agents, the per-step complexity of GAT aggregation is $O{=}N^2 \bullet d$, where d denotes the feature dimension; in practice, N remains small and manageable. The overall training complexity is $O = T \bullet B \bullet (N \bullet d^2))$, with T representing the number of iterations and B the batch size. The space complexity involves approximately 100,000 parameters, requiring around 500 MB of memory usage.

The parameters are shown in Tab. F2.

Table F2
Training hyperparameters.

Parameter types	Value	Unit
Fully connected network	3	layer
Activation function	ReLU	/
Actor Learning Rate	0.001	/
Critic learning rate	0.002	/
Experience replay buffer	10,000	transitions
Batch size	128	/
Graph Attention Layer (GAT)	2	layer
Training cycle	20,000	episodes
Unit penalty cost coefficient λ	0.5	/
Reward function parameters α	0.30	/
Reward function parameters γ	0.20	/
Reward function parameters μ	0.45	/
Reward function parameters ρ	0.15	/

Key implementation features:

1) Softmax constraint: The Actor output layer uses Softmax to ensure that actions satisfy the probabilistic simplex constraint $\left(\sum_{i=1}^{N} p_i = 1, p_i \geqslant 0; \sum_{i=1}^{N} q_i = 1, q_i \geqslant 0\right)$.

2) Policy embedding: The load-side Critic network is explicitly embedded into the power-side Actor network as a policy function to solve the "expected response" problem in sequential decision-making.

The experiment evaluated the following key metrics:

1) Global Profit (GP): The overall revenue of the VPP, calculated as the profit on the power supply side minus the cost on the load side and the penalty for power imbalance:

$$GP = \sum profit(\pi_i) - \sum cost(\pi_i) \tag{F1}$$

2) Default Risk (DR): Defined as the probability that the power imbalance exceeds 5% of the rated power threshold, calculated through Monte Carlo simulation of 100 disturbance scenarios:

$$DR = \left(\frac{\sum_{i=1}^{N} I(|\Delta P_i| > \theta)}{N}\right) \times 100\% \tag{F2}$$

3) Algorithm convergence speed: Measured by the number of iterations required for the reward function to converge to a stable value:

$$CS = \min\{t || R(t) - R(t-1) | < \varepsilon \forall k \text{steps} \} \tag{F3}$$

4) Robustness: Profit volatility under uncertainty is assessed using standard deviation.

$$Ro = \sqrt{\frac{\sum_{i=1}^{N} (GP_i - \mu)^2}{N-1}} \tag{F4}$$

5) Equilibrium Quality Index (EI): The smaller the value, the closer it is to the subgame Nash equilibrium. Zero is the best value.

$$EI = \frac{2}{N-1} \sum_{i=1}^{N-1} \left(\frac{profit(\pi_i) - profit(\pi_i)}{profit(\pi_i) + profit(\pi_i)} + \frac{cost(\pi_i) - cost(\pi_i)}{cost(\pi_i) + cost(\pi_i)} \right), EI \in [0, 1] \tag{F5}$$

Case selection and data sources:

Two representative cases were selected: (1) Suzhou Industrial Park VPP pilot: Located in the Yangtze River Delta region, rich in resource types, including distributed photovoltaic, energy storage, and industrial load. (2) Xuancheng VPP pilot area in Anhui Province: Focusing on electric vehicle clusters and renewable energy integration.

Data source: The data for Suzhou Industrial Park comes from the "Analysis Report on the Development Status of Virtual Power Plants in the Yangtze River Delta" (2024). The data for Anhui Province is based on the column "2024 China Virtual Power Plant Industry Research Report". The parameter Tab. F3 are as follows:

Table. F3
. Virtual power plant core parameters.

Data types	(Su) Value	Unit	(An) Value	Unit
Total scale of aggregated resources	14.62	MW	487.9	MW
Distributed photovoltaic	13.60	MW	121.3	MW
Industrial load baseline	14.62	MW	350.8	MW
Adjustable range	[-25%,25%]	/	[-30%,30%]	/
Energy storage unit	1.06	MW	5	MW
Electric vehicle cluster capacity	0.39	MW	9.9	MW
Grid-connected adjustable load	7.47	MW	113.2	MW

Uncertainty factors are derived from meteorological data adapted from historical data of the China Meteorological Administration in the summer of 2023, with photovoltaic power output fluctuating by 10%-15%; load disturbances are based on the IEEE dataset and adjusted to the Chinese industrial load model using Poisson impact, with an average interval of 4 h and an amplitude of ± 15 kW; electricity price data are from the National Energy Administration database, with a base price of 0.6 yuan/kWh and a fluctuation range of [0.4, 0.9] yuan/kWh. Historical operating data are derived from actual profits and default events from the 2023 operating indicators summarized in the report. The simulation environment uses MATLAB/Simulink combined with Pytorch, with a time span of 24 h and 24 time periods. The MADRL algorithm uses optimized parameters.

The analysis process consists of three steps: (1) Data preprocessing and uncertainty modeling: Import real historical data and add random disturbances to simulate the actual environment. (2) Model application: Embed the SG-MADDPG framework into the case system, with the upper layer using sequential game optimization of the electricity price hybrid strategy and the lower layer using multi-agent reinforcement learning to schedule specific units. (3) Performance Evaluation: Calculate key performance indicators and compare them with actual operating benchmarks. Uncertainty modeling is the same as in the experimental section above. In the SG-MADDPG application, the power supply side acts as the leader to formulate the electricity pricing strategy p, and the load side acts as the follower to adjust q. DRO ensures convergence to Nash equilibrium.

Data availability

Data will be made available on request.

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